A Project Report on

Prakriti: Human Body-type Identification to Recommend Lifestyle using IoT and ML

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

Suraj Vishwakarma: 22106019

Shravan Thakur: 22106086

Sameer Singh: 22106116

Ritesh Singh: 22106037

Under the Guidance of

Dr. Mahesh Pawaskar

Department of Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

UNIVERSITY OF MUMBAI

2025-2026

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "Prakriti: Human Body-type Identification to Recommend Lifestyle using IoT and ML" is a bonafide work of **Suraj Vishwakarma(22106019), Shravan Thakur(22106086), Sameer Singh(22106116), Ritesh Singh(22106037)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

Dr. Mahesh Pawaskar
Guide

Prof. Sayali Badhan
Project Co-ordinator

Dr. Jaya Gupta
Head of Department

Dr. Uttam Kolekar
Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for B.E.

This project report entitled ***Prakriti: Human Body-type Identification to Recommend Lifestyle using IoT and ML*** by ***Suraj Vishwakarma, Shravan Thakur, Sameer Singh, Ritesh Singh*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning)***, ***2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Suraj Vishwakarma 22106019

Shravan Thakur 22106086

Sameer Singh 22106116

Ritesh Singh 22106037

Date:

Abstract

Prakriti is an innovative health analysis platform that integrates ancient Ayurvedic principles with modern IoT and Machine Learning technologies to provide personalized lifestyle recommendations. The system identifies an individual's Dosha type Vata, Pitta, or Kapha using real-time biometric data collected through IoT sensors such as ECG, Pulse, Temperature, and Oximeter sensors. The data is processed using machine learning algorithms (Random Forest, TensorFlow, Scikit-learn) to classify body types and generate customized recommendations for diet, yoga, meditation, and lifestyle. Unlike conventional health monitoring systems that only provide raw biometric values, Prakriti interprets this data through the lens of Ayurveda, offering actionable lifestyle changes rooted in holistic well-being. The project bridges the gap between traditional healthcare and modern preventive medicine by offering proactive wellness solutions rather than reactive treatment. It democratizes access to Ayurvedic wisdom by embedding it in a user-friendly platform powered by IoT devices and cloud-based infrastructure. Prakriti supports the United Nations Sustainable Development Goals: SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation, and Infrastructure), SDG 10 (Reduced Inequalities), and SDG 17 (Partnerships for the Goals), by fostering innovation, promoting wellness, and democratizing access to equitable healthcare.

Keywords: *Ayurveda, IoT, Machine Learning, Dosha, Prakriti, Preventive Healthcare, Personalized Wellness*.

Contents

1. Introduction 1
1.1 The Need for Preventive Healthcare 1
1.2 The Wisdom of Ayurveda 2
1.3 The Gap Between Modern Healthcare and Tradition 2
1.4 The Role of IoT and AI/ML in Healthcare 3
1.5 Towards an Intelligent Preventive Healthcare Ecosystem 4
1.6 Digital Transformation of Traditional Healthcare Knowledge 4
1.7 Vision for a Smart Preventive Healthcare System 5
2. Survey 6
3. Limitation of Existing System 17
4. Problem Statement and Objective 19
4.1 Problem Statement 19
4.2 Objectives 19
4.3 Scope 20
5. Proposed System 21
5.1 Architecture of Proposed System 21
5.2 Data Flow Diagram 24
5.3 Use Case Diagram: 26
5.4 Sequence Diagram 28
5.5 Activity Diagram: 29
5.6 Key Components of the Proposed Framework: 33
6. Experimental Setup 35
6.1 Software Requirements: 36
6.2 Hardware Requirements: 36
7. Results and Discussion 38
8. Conclusion and Future Work 48
8.1 Conclusion 48
8.2 Future Work 49
References 51
Appendix 53
Publication 55

LIST OF FIGURES

Figure 5.1 Architecture Diagram 23
Figure 5.2.1 Data Flow Diagram 25
Figure 5.2.2 Data Flow Diagram 26
Figure 5.3 Use Case Diagram 27
Figure 5.4 Sequence Diagram 29
Figure 5.5 Activity Diagram 33
Figure 5.6 Flow Diagram 34
Figure 7.1 Vaidya Home Page 40
Figure 7.2 Dosha Prediction Model 41
Figure 7.3 Dosha Prediction and Recommendation 42
Figure 7.4 Real Time Monitoring Dashboard 43
Figure 7.5 AI Chatbot 43
Figure 7.6 Nearby Ayurvedic Doctor Appointment page 44
Figure 7.7 IoT devices to Cloud data transfer 45

LIST OF TABLES

Table 2.1: Literature Survey……...13
Table 7.1.1: Comparative Advantage over Traditional Systems……..... 41
Table 7.1.2: Final Accuracy Table by Model and Target……46
Table 7.1.3: Ensemble Learning Accuracy……47

LIST OF GRAPHS

Graph 7.1.1: Comparative Analysis of ML Models……...……46
Graph 7.1.2: Ensemble Model Performance Comparison…….……..... 47

ABBREVAION

AI	*Artificial Intelligence*
ML	*Machine Learning*
IoT	*Internet of Things*
HR	*Heart Rate*
ECG	*Electrocardiogram*
SpO_2	*Peripheral Capillary Oxygen Saturation*
ANN	*Artificial Neural Network*
CNN	*Convolution Neural Network*
SVM	*Support Vector Machine*
WHO	*World Health Organization*
HTTP	*Hyper Text Transfer Protocol*
SMS	*Short message service*
HMS	*Health Monitoring System*
CMI	*Criticality Measure Index*
MQTT	*Message Queuing Telemetry Transport*
GSM	*Global System for Mobile Communication*
SVC	*Support Vector Classification*
USB	*Universal Serial Bus*
CSS	*Cascading Style Sheets*
RASPRO	*Rapid Active Summarization for Effective Prognosis*

Chapter 1

Introduction

1. Introduction

1.1 The Need for Preventive Healthcare

In the modern era, health challenges are evolving rapidly due to the increasing demands of lifestyle and work-related stress. With long working hours, sedentary habits, irregular sleep cycles, and unbalanced dietary practices, individuals often ignore the early signs of health imbalance. While minor symptoms may seem harmless initially, they gradually accumulate into significant health complications. Unfortunately, most people wait until the manifestation of illness before seeking medical intervention, leading to a reactive approach to healthcare.

This reliance on curative medicine contributes to a rise in chronic lifestyle-related diseases such as hypertension, diabetes, cardiovascular problems, and stress-induced disorders. Reports from the World Health Organization (WHO) suggest that nearly 70% of global deaths are caused by non-communicable diseases, most of which could be prevented through timely lifestyle adjustments. Despite the availability of modern healthcare facilities, preventive care remains underutilized, leaving a gap in health awareness and long-term well-being.

Another challenge lies in the absence of continuous and personalized health monitoring. While health check-ups are conducted at intervals, they provide only a snapshot of one's current condition rather than a comprehensive picture of daily health fluctuations. This lack of real-time insight limits an individual's ability to detect imbalances early, often resulting in delayed treatment and reduced quality of life. Thus, there is an urgent need for healthcare systems that empower individuals to monitor, understand, and manage their health proactively.

1.2 The Wisdom of Ayurveda

Ayurveda, the ancient Indian system of medicine, offers a comprehensive and preventive approach to health and well-being. With a history spanning over 5,000 years, Ayurveda emphasizes balance, harmony, and holistic living. At the heart of Ayurvedic practice lies the concept of Prakriti—an individual's unique constitution determined by the three Doshas: Vata, Pitta, and Kapha. Each person is born with a distinct combination of these Doshas, which governs their physical, emotional, and behavioral tendencies. Maintaining equilibrium among them is considered essential for optimal health and vitality.

Unlike modern healthcare systems that primarily address symptoms, Ayurveda focuses on identifying root causes of imbalance. It prescribes personalized lifestyle practices, dietary patterns, herbal remedies, and daily routines that align with one's Prakriti. For instance, a Pitta-dominant individual may be advised to adopt cooling foods, stress-reducing practices, and meditation techniques to avoid inflammatory conditions. This personalized and preventive orientation makes Ayurveda a powerful tool for long-term wellness management.

However, despite its holistic benefits, the practical application of Ayurveda faces limitations. Accurate Dosha assessment often requires expert consultation with trained practitioners, which may not be easily accessible or affordable for the majority of people. Additionally, the integration of Ayurvedic recommendations into daily routines without professional guidance can be challenging for individuals unfamiliar with its principles. As a result, the preventive advantages of Ayurveda remain underutilized, especially in urban societies that need them the most.

1.3 The Gap Between Modern Healthcare and Tradition

Technological innovations have significantly advanced healthcare in recent years. The proliferation of wearable devices, mobile health applications, and remote monitoring tools has

enabled continuous tracking of biomedical indicators such as heart rate, oxygen saturation, and physical activity. These tools have improved awareness and encouraged individuals to take greater responsibility for their health. However, their scope remains restricted to Western biomedical metrics, focusing more on physical parameters than holistic well-being.

Moreover, most of these devices provide raw data without meaningful interpretation. A step count or heart rate measurement, for instance, does not directly translate into actionable lifestyle changes unless contextualized. This creates a gap between data availability and actual health outcomes. While modern healthcare excels in precision and early detection of diseases, it lacks the interpretive wisdom that traditional systems like Ayurveda bring in connecting health data to lifestyle recommendations.

This gap is further widened by the absence of platforms that integrate ancient knowledge systems with digital technology. Despite Ayurveda's global recognition as a holistic healing framework, few digital solutions have attempted to merge its insights with real-time health monitoring. Therefore, there exists a unique opportunity to design a solution that not only tracks biometric data but also interprets it through Ayurvedic principles to provide actionable and personalized wellness guidance.

1.4 The Role of IoT and AI/ML in Healthcare

The rapid evolution of Internet of Things (IoT) technology has made it possible to continuously monitor a wide range of biometric parameters using portable sensors and wearable devices. Devices embedded with sensors can track vital signs such as heart rate, pulse, oxygen levels, temperature, and sleep patterns in real time. This data, when analysed systematically, provides valuable insights into an individual's health trends and potential risk factors. IoT-driven solutions are becoming indispensable in creating personalized healthcare ecosystems.

Artificial Intelligence (AI) and Machine Learning (ML) add another dimension to this transformation. By analysing vast amounts of data, ML models can identify patterns, predict outcomes, and generate insights that may not be visible through manual analysis. For instance, ML algorithms can classify individuals based on health patterns, detect subtle deviations from normal ranges, and even forecast potential imbalances before symptoms appear. This predictive capability aligns seamlessly with Ayurveda's preventive philosophy, creating a synergy between ancient wisdom and modern technology.

When combined, IoT and AI/ML hold the potential to revolutionize healthcare delivery. They enable continuous data collection, intelligent processing, and personalized recommendations all of which are critical for shifting from reactive treatment to proactive health management. Integrating these technologies with Ayurvedic principles creates a framework where modern precision meets traditional interpretation, leading to more meaningful and holistic health interventions.

1.5 Towards an Intelligent Preventive Healthcare Ecosystem

In order to bridge the gap between traditional wisdom and modern technological capabilities, there is a growing need for intelligent healthcare systems that combine real-time monitoring with personalized wellness guidance. Such systems can empower individuals to become active participants in managing their health rather than passive recipients of treatment. By integrating biometric data collection with intelligent analysis, it becomes possible to identify patterns that indicate early signs of imbalance in the body. When these insights are interpreted through a holistic framework such as Ayurveda, individuals can receive lifestyle recommendations that align with their unique physiological constitution. This approach not only enhances awareness but also encourages sustainable behavioural changes that support long-term well-being.

1.6 Digital Transformation of Traditional Healthcare Knowledge

The digital transformation of healthcare provides an opportunity to revive and modernize ancient medical knowledge systems. By translating Ayurvedic principles into computational models, it becomes possible to automate Dosha analysis and health assessment in a structured and accessible manner. Machine learning algorithms can analyse physiological signals and behavioural patterns to estimate Dosha tendencies and detect deviations from an individual's natural balance. Furthermore, mobile and web-based platforms can deliver personalized recommendations related to diet, exercise, sleep patterns, and stress management based on these insights. Such digital solutions can make Ayurvedic guidance more accessible to a broader population, particularly in urban environments where traditional consultations may not be easily available.

1.7 Vision for a Smart Preventive Healthcare System

The convergence of Ayurveda with modern technologies such as the Internet of Things (IoT) and Artificial Intelligence/Machine Learning (AI/ML) provides a promising pathway for the development of intelligent preventive healthcare systems. By utilizing wearable devices and sensor-based technologies, it becomes possible to continuously collect important biometric data such as heart rate, body temperature, sleep patterns, and physical activity levels. Unlike traditional health check-ups that provide only occasional insights, these technologies enable real-time monitoring of an individual's physiological condition. When this data is processed using intelligent algorithms, meaningful patterns and trends can be identified, allowing the system to detect subtle deviations from normal health conditions at an early stage. Such proactive monitoring empowers individuals to take timely actions before minor imbalances evolve into serious health issues.

Furthermore, the insights generated from this data can be interpreted through the holistic framework of Ayurvedic principles. By mapping physiological patterns to Dosha characteristics—Vata, Pitta, and Kapha—the system can generate personalized lifestyle and wellness recommendations tailored to an individual's constitution. These recommendations may include dietary suggestions, stress management practices, physical activities, and daily routines that help restore and maintain Dosha balance. The ultimate objective of such an integrated platform is to create a user-centric health management system that not only tracks physical parameters but also promotes balanced and mindful living. By combining the analytical power of modern technology with the wisdom of traditional medicine, this approach has the potential to transform preventive healthcare into a more proactive, accessible, and holistic model of well-being.

Chapter 2

Literature Survey

2. Survey

1. Remote Health Monitoring System using IoT: A Review

The review paper "Remote Health Monitoring System using IoT: A Review" (2025) presents an overview of IoT-based healthcare monitoring systems. It explains how these systems use sensors, Wi-Fi modules, and cloud platforms to track vital signs like heart rate, blood pressure, and SpO_2 in real time. Unlike traditional hospital monitoring, IoT enables continuous remote observation by doctors and caregivers. The study highlights technologies such as ESP8266, MAX30100 sensors, and cloud services like AWS and Azure. It also discusses the use of mobile apps (e.g., Blynk) for easy patient monitoring. Advanced concepts like AI and edge computing improve predictive analysis and response time. Blockchain is noted for enhancing data security. The paper emphasizes benefits in elderly care, chronic disease management, and pandemic situations. However, challenges include data privacy risks, high costs, and poor rural connectivity. Overall, it concludes that IoT has strong potential to transform healthcare into a more accessible and efficient system.

2. **Exploring the Ayurvedic Concept of Body Types and Their Relevance to Heart Rate Variability (HRV)**

The paper "Exploring the Ayurvedic Concept of Body Types and Their Relevance to Heart Rate Variability (HRV)" (2025) studies the link between Ayurvedic doshas and HRV. Ayurveda classifies individuals into Vata, Pitta, and Kapha based on physical and mental traits. The research uses signal processing techniques like Phase Space Reconstruction (PSR) and Approximate Entropy (ApEn) to analyse HRV patterns. HRV is an important indicator of autonomic nervous system activity and heart health. The results show that Vata types have higher variability and irregular heart rhythms. Kapha types display stable and regular patterns, while Pitta types show moderate variability. The study also suggests using wearable PPG sensors for continuous HRV monitoring. This enables real-time assessment of body constitution in daily life. The research combines traditional Ayurveda with modern technology. It highlights the potential for personalized healthcare and digital health monitoring using HRV analysis.

3. **Personalized Health Recommendation System with Machine Learning**

The paper "Personalized Health Recommendation System with Machine Learning" (2025) presents an ML-based system for personalized healthcare advice. It collects user data such as symptoms, age, gender, medical history, and lifestyle through a web interface. The data is pre-processed using normalization, encoding, and handling missing values to improve accuracy. A Support Vector Classification (SVC) model is used to predict possible diseases based on symptoms. After prediction, the system provides recommendations including medications, dosage, diet, and lifestyle changes. These suggestions are retrieved from a medical database. Unlike traditional systems, it offers personalized and adaptive recommendations. The model also improves over time using user feedback. The study highlights its usefulness in remote healthcare services. It can reduce hospital workload and support preventive care. Overall, the system enhances accessibility and efficiency in healthcare delivery.

4. **Personalized Healthcare Web-Application**

The paper "Personalised Healthcare Web-Application" (2025) presents an AI-based system for personalized healthcare recommendations. It uses AI, machine learning, and big data to analyze patient data such as symptoms, medical history, and genetic information. Data is collected from wearables, EHRs, and mobile health apps. The system identifies patterns to predict health risks and suggest treatments and lifestyle changes. IoT technology enables continuous monitoring of patient vitals. Cloud computing supports large-scale data processing and system scalability. The

system helps improve early disease detection and clinical decision-making. It also reduces hospital visits and overall healthcare costs. However, challenges like data privacy, security, and ethical concerns remain. Technologies like encryption and blockchain are suggested for secure data handling. The study concludes that future advancements will enhance reliability and accessibility of such systems.

5. **Wearable Sensors Based on Artificial Intelligence Models for Human Activity Recognition**

The paper "Wearable Sensors Based on Artificial Intelligence Models for Human Activity Recognition" (2024) presents an AI-based system for recognizing human activities. It uses data from wearable sensors like accelerometers, gyroscopes, and barometers. The system can detect activities such as walking, running, standing, and falling. Both traditional ML models (SVM, Random Forest, K-NN) and deep learning (CNN) are used. A sensor-specific CNN with a late-fusion technique improves accuracy. Each sensor is processed separately, and results are combined for better prediction. The model achieves 94.83% accuracy, outperforming SVM (83.10%). The study shows that multi-sensor data improves reliability. Applications include healthcare monitoring, fall detection, and sports tracking. However, challenges include limited datasets and sensor placement issues. Overall, the system enhances activity recognition using AI and wearable technology.

6. **Enhancing Ayurvedic Diagnosis using Multinomial Naïve Bayes and K-Modes Clustering**

The paper "Enhancing Ayurvedic Diagnosis using Multinomial Naïve Bayes and K-Modes Clustering" presents an ML-based approach for Prakriti classification. Ayurveda traditionally classifies individuals into three doshas—Vata, Pitta, and Kapha. This study expands the classification to seven categories to include overlapping doshas. It uses a dataset with 147 categorical features related to body characteristics. Chi-Square feature selection is applied to identify important attributes. K-Modes clustering is used to group categorical data effectively. For classification, Multinomial Naïve Bayes and Decision Tree models are used. Results show that Naïve Bayes performs better with around 90% accuracy. The model also achieves high precision, recall, and F-score. The approach improves the accuracy of Ayurvedic diagnosis. It demonstrates how ML can modernize traditional healthcare systems.

7. **Design and Implementation of Advanced Real-Time Patient Health Monitoring System Using IoT**

The paper "Design and Implementation of Advanced Real-Time Patient Health Monitoring System Using IoT" (2024) presents an IoT-based system for continuous health monitoring. It uses a Raspberry Pi with sensors to measure heart rate, temperature, SpO_2, respiration, and movement. Data is collected in real time and transmitted using protocols like MQTT, HTTP, and ThingSpeak. Healthcare providers can access this data remotely through cloud platforms. The system sends alerts via GSM (SMS) when abnormal conditions are detected. A Pi camera and mobile/web apps enable live monitoring and feedback. It improves patient comfort by reducing manual measurements. The system also supports trend analysis for better healthcare decisions. Challenges include cost, sensor accuracy, and rural connectivity. The study highlights future integration with predictive analytics. Overall, it enhances real-time and personalized healthcare delivery.

8. **IoT-Enabled Health Monitoring Systems: Transforming Healthcare Delivery**

The paper "IoT-Enabled Health Monitoring Systems: Transforming Healthcare Delivery" (2024) explores the impact of IoT in healthcare. It presents a system that uses wearable sensors, ECG modules, and temperature sensors for continuous monitoring. Microcontrollers like ATmega328P and Raspberry Pi are used to process data. The data is transmitted to cloud platforms using IoT protocols. Advanced analytics and machine learning provide real-time insights to doctors. The system improves accessibility and reduces healthcare costs. It enables timely interventions and better patient outcomes. Integration with smartphones allows easy access to health data. However, challenges include security, privacy, and scalability issues. Overall, the study highlights IoT's potential in personalized and efficient healthcare systems.

9. **Smart Healthcare Monitoring System Using IoT**

The paper "Smart Healthcare Monitoring System Using IoT" (2023) presents a real-time health monitoring system using IoT. It uses sensors like DS18B20 for temperature and MAX30100 for heart rate and SpO_2. A Raspberry Pi 4B is used to process and manage the data. A 4G GSM/GPRS module enables location tracking and emergency SMS alerts. Data is stored in a MySQL cloud database for easy access. A mobile app built with Flutter and Dart provides real-time monitoring and visualization. Doctors can view patient data remotely and make quick decisions. The system shows high accuracy, with over 97% precision. It reduces hospital visits

and improves emergency response. Challenges include scalability and adding more health parameters. Overall, it offers a reliable and patient-friendly healthcare solution.

10. An IoT-Based Framework for Personalized Health Assessment and Recommendations Using Machine Learning

The paper "An IoT-Based Framework for Personalized Health Assessment and Recommendations Using Machine Learning" (2023) proposes a smart healthcare system combining IoT and ML. It collects real-time data like pulse rate, SpO_2, temperature, and BMI using sensors and Arduino. The data is analysed using ML models such as Random Forest, Cat Boost, Logistic Regression, and MLP. A large dataset of around 984,000 samples is used for training and testing. The Cat Boost model achieves nearly 100% accuracy, outperforming others. The system classifies users into health categories like fit, medium risk, and high risk. Based on results, it provides personalized diet and fitness recommendations. A Flask-based web interface is used for user interaction. The system supports continuous monitoring and preventive healthcare. It helps improve personalized wellness management. Overall, it shows the effectiveness of IoT and ML in smart healthcare systems.

11. Machine Learning for Healthcare Wearable Devices: The Big Picture

The paper "Machine Learning for Healthcare Wearable Devices: The Big Picture" (2022) reviews the use of ML in wearable healthcare systems. It explains that devices like smartwatches, chest straps, and rings are part of IoMT. These devices monitor signals such as heart rate, ECG, temperature, and movement. The collected data is analyzed using machine learning algorithms. Applications include activity recognition, fall detection, stress and sleep monitoring, and disease diagnosis. The paper discusses methods like supervised learning, clustering, and deep learning. It also highlights challenges such as limited data, power consumption, and storage issues. Privacy and security concerns are also important factors. The study emphasizes the need for reliable data and model accuracy. Overall, it shows that ML with wearables can enable real-time and personalized healthcare.

12. Patient Health Monitoring System Using IoT

The paper "Patient Health Monitoring System Using IoT" (2022) presents a smart edge-based healthcare monitoring system. It uses wearable sensors to measure parameters like heartbeat, ECG, temperature, SpO_2, and fall detection. The system is designed for rural and low-resource settings. It uses microcontrollers with GSM and Wi-Fi for data transmission. Two key software

engines, RASPRO and CMI, summarize data and generate alerts. Important health updates are sent directly to doctors in real time. Detailed data can also be accessed through the cloud. The system improves responsiveness and monitoring accuracy. It reduces hospital visits and supports timely medical intervention. Challenges include scalability and data security. Overall, it enhances accessible and efficient healthcare delivery.

13. Machine Learning Based Classification of Ayurvedic Constitution Types

The paper "Recapitulation of Ayurveda Constitution Types by Machine Learning of Phenotypic Traits" (2017) presents an ML-based approach for Prakriti classification. Ayurveda classifies individuals into Vata, Pitta, and Kapha types. Traditionally, this is done through subjective clinical assessment. The study uses phenotypic data from 147 individuals. Both unsupervised and supervised machine learning methods are applied. Clustering techniques group individuals into three main Prakriti types. Supervised models accurately classify individuals based on selected features. The results show that Prakriti types can be represented as measurable data patterns. This makes the classification more objective and scientific. The study bridges traditional Ayurveda with modern data science. It highlights the role of ML in personalized healthcare.

14. Acquisition and Quantitative Analysis of the Arterial Pulse

The thesis "Acquisition and Quantitative Analysis of the Arterial Pulse" (2011) presents a scientific approach to pulse-based diagnosis. It introduces Nadi Tarangini, a low-cost system for capturing accurate pulse signals. Unlike traditional subjective methods, it uses signal processing techniques for analysis. Methods include PSWT-DTW, multifractal analysis, and recurrence plots. A Hybrid SVM classifier is used for better diagnosis accuracy. The study uses a large dataset of pulse signals from many subjects. It also compares results with ECG datasets like MIT-BIH. Results show improved accuracy and reduced false negatives. The system can detect variations across age and diseases. Challenges include data imbalance and lack of standardization. Overall, it supports integrating traditional Ayurveda with modern technology.

15. Nadi Tarangini: A Pulse Based Diagnostic System

The paper "Nadi Tarangini: A Pulse Based Diagnostic System" (2007) presents a computer-aided system for Ayurvedic pulse analysis. It uses pressure sensors, strain gauges, and a 16-bit digitizer to capture accurate pulse signals. Three sensors are placed on the wrist to record Vata, Pitta, and Kapha pulses. The system provides high-quality data with minimal noise. It shows

consistent and reproducible results across sessions. The study also observes age-related variations in pulse signals. The captured waveforms have detailed features and secondary peaks. These features improve diagnostic interpretation. The system can be combined with machine learning for classification. It bridges traditional Ayurveda with modern technology. Overall, it offers a low-cost and scalable diagnostic solution.

Year	Title	Domain	Algorithm	Dataset	Gap
2025	Remote Health Monitoring System using IoT: A Review	IoT + Healthcare	Review-based (IoT + ML concepts)	No dataset (survey)	Standardization, scalability, privacy, weak AI integration
2025	Exploring the Ayurvedic Concept of Body Types and Their Relevance to Heart Rate Variability (HRV)	Ayurveda + Biomedical Signal Processing	Phase Space Reconstruction (PSR), Approximate Entropy (ApEn)	HRV signals collected from physiological monitoring	Requires larger clinical datasets and validation for wearable deployment
2025	Personalized Health Recommendation System with Machine Learning	Machine Learning + Healthcare	Support Vector Classification (SVC)	Symptom-based medical dataset	Prediction accuracy depends on user symptom input and dataset diversity
2025	Personalised Healthcare Web-Application	AI + IoT + Healthcare	Machine Learning models + Big Data analytics	Data from wearables, EHRs, and mobile health applications	Data privacy, ethical concerns, and system interoperability challenges
2024	Wearable Sensors Based on Artificial Intelligence Models for Human Activity Recognition	Wearable Sensors + AI + Healthcare	CNN, SVM, Random Forest, K-NN	Multi-sensor dataset (triaxial accelerometer, gyroscope, barometer)	Limited dataset diversity, sensor placement variability, generalization issues in real-world environments

2024	Enhancing Ayurvedic Diagnosis using Multinomial Naïve Bayes and K-Modes Clustering	Ayurveda + Machine Learning	Multinomial Naïve Bayes, K-Modes Clustering, Decision Tree	Dataset with 147 categorical Ayurvedic body features	Dosha overlapping complexity and need for larger real-world datasets
2024	Design and Implementation of Advanced Real-Time Patient Health Monitoring System Using IoT	IoT + Healthcare	Raspberry Pi, sensors, MQTT, GSM	Prototype sensor data	Limited scalability, no AI
2024	IoT-Enabled Health Monitoring Systems: Transforming Healthcare Delivery	IoT + Healthcare	Sensors (HR, Temp, ECG), Raspberry Pi, IoT cloud, ML analytics	Prototype sensor data	Security, privacy, scalability issues
2023	Smart Healthcare Monitoring System Using IoT	IoT + Healthcare	Raspberry Pi 4B, DS18B20, MAX30100, SIM7600E, Python, Flutter app, MySQL	Tested on 30 persons (temp, HR, SpO_2)	Limited parameters, small sample size, no AI decision support
2023	An IoT-Based Framework for Personalized Health Assessment and Recommendations Using Machine Learning	IoT + Healthcare + Machine Learning	Random Forest, CatBoost, Logistic Regression, MLP	IoT sensor dataset (~984,000 samples) including pulse, SpO_2, BMI, temperature	Limited health parameters and need for stronger privacy/security mechanisms

2022	Machine Learning for Healthcare Wearable Devices: The Big Picture	Wearable Healthcare + Machine Learning	Supervised Learning, Clustering, Deep Learning	Wearable sensor data (heart rate, ECG, EDA, body temperature, movement data)	Limited data availability, power consumption constraints, storage limitations, privacy and security concerns
2022	Patient Health Monitoring System Using IoT	IoT + Healthcare	Wearable sensors, Arduino, Wi-Fi, GSM, cloud, RASPRO + CMI engines	Prototype patient data	Limited scope (ECG, temp, SPO_2), single patient, no advanced ML
2017	Recapitulation of Ayurveda Constitution Types by Machine Learning of Phenotypic Traits	Ayurveda + Machine Learning	Unsupervised Clustering, Supervised Classification	Phenotypic dataset of 147 individuals from North Indian population	Small dataset and limited population diversity
2011	Acquisition and Quantitative Analysis of the Arterial Pulse	Biomedical signals, Ayurveda, CSE	PSWT-DTW, Hybrid-SVM, fractal & chaos methods	MIT-BIH ECG DB, In-house pulse DB (*Nadi Tarangini*)	Limited quantitative studies, data imbalance, high variation, weak standardization vs ECG

2007	Nadi Tarangini: A Pulse Based Diagnostic System	Ayurveda + IoT/Signal Processing in Healthcare	Pressure sensors, waveform analysis, wavelet denoising	31 patients, pulse waveform data	Small dataset, no ML, lacks large-scale validation

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Current health monitoring and wellness systems suffer from several limitations that reduce their effectiveness:

- **Reactive Approach** – Most existing health monitoring systems focus primarily on providing responses only after a health issue has already occurred. They lack strong mechanisms for preventive care and early detection, which are essential for identifying imbalances before they turn into major problems.
- **Limited Personalization** – Many current wellness applications provide generic, one-size-fits-all recommendations that do not account for an individual's unique body constitution (Prakriti), lifestyle habits, or environmental influences. As a result, users receive advice that may not be truly effective or suitable for their personal health needs.
- **Lack of Integration with Traditional Medicine** – Existing IoT-based health solutions depend heavily on biomedical parameters and tend to ignore holistic frameworks like Ayurveda. This creates a gap, as Ayurveda emphasizes balance between body, mind, and environment, which is crucial for long-term wellness and preventive healthcare.

- **Data Fragmentation** – Health-related information is often spread across multiple platforms such as sensors, mobile apps, and wearable fitness devices. This fragmentation makes it difficult to consolidate data into a unified view, preventing comprehensive analysis of an individual's overall wellness.
- **Accessibility Issues** – Advanced healthcare systems often demand costly infrastructure, specialized equipment, or expert consultation. This makes them unaffordable and inaccessible for large sections of the population, particularly in rural or underprivileged areas.
- **Low Preventive Capabilities** – Without the use of predictive analytics and continuous long term data analysis, many existing systems fail to detect subtle health variations or early warning signs. This limits their ability to identify risks at an early stage and provide timely preventive interventions.
- **Real-Time Monitoring Challenges** – Although many devices claim to offer continuous tracking, existing systems often struggle with accuracy, delayed responses, or poor synchronization of sensor data, reducing the reliability of real-time health monitoring.
- **Poor User Engagement and Interface Design** – A significant limitation of current solutions is their complex or non-intuitive dashboards, which make it difficult for ordinary users to interpret health insights. As a result, many individuals do not actively engage with the system. • Limited AI-Driven Insights – While some systems use AI, most are restricted to basic trend detection and lack deeper analytical capabilities, such as combining biomedical data with lifestyle and traditional health frameworks. This limits the system's ability to provide holistic, personalized recommendations.

Prakriti aims to overcome these limitations by integrating continuous IoT-based biometric monitoring with machine learning models grounded in Ayurvedic principles, delivering personalized, preventive, and holistic health recommendations.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1 Problem Statement

In today's fast-paced world, people often miss early signs of health issues due to the lack of real time monitoring and personalized guidance. While Ayurveda offers holistic wellness solutions, identifying one's Dosha and corresponding remedies without expert help is challenging. This project integrates IoT sensors with a machine learning model to monitor health metrics, classify Dosha, and provide personalized Ayurvedic recommendations, making preventive healthcare more accessible and effective.

4.2 Objectives

- The system aims to develop a real-time health monitoring framework using advanced sensors that continuously track vital parameters such as heart rate, temperature, oxygen level, and blood pressure. This ensures timely detection of abnormalities and supports proactive healthcare management.

- The platform will be designed to generate automated health reports in PDF format, making them easily accessible for patients as well as healthcare providers. These reports can be stored for long-term record-keeping and shared conveniently during consultations.
- Based on the collected health data, the system will offer personalized diet and exercise recommendations tailored to the individual's unique requirements. This promotes healthier lifestyle choices and aids in preventing potential health complications.
- A hybrid consultation model will be established that combines AI-driven analysis with expert insights from human doctors. This approach ensures both accuracy in initial assessments and reliability through professional medical validation.
- The solution will include a modern and user-friendly dashboard designed for effective visualization and interpretation of health data. This interactive interface will help users easily track their wellness trends and make informed health decisions.
- By integrating Ayurvedic knowledge with AI technology, the system aims to deliver holistic and preventive healthcare solutions. This fusion of traditional wisdom with modern innovation ensures a balanced approach to maintaining physical and mental well-being.

4.3 Scope

- The project enables continuous real-time monitoring of vital health parameters using sensors to provide accurate and timely health insights.
- It generates automated health reports in PDF format, ensuring that users and healthcare providers have structured documentation of health records.
- The system offers personalized diet and exercise recommendations tailored to individual health conditions and Ayurvedic body types.
- It provides a hybrid consultation model that combines AI-driven diagnostic suggestions with expert advice from human doctors.
- The project includes a modern, user-friendly dashboard that allows users to visualize their health data effectively and track improvements over time.
- It integrates Ayurvedic knowledge with AI technology to deliver holistic healthcare solutions that bridge traditional wisdom and modern innovation.

Chapter 5

Proposed System

5. Proposed System

5.1 Architecture of Proposed System

The system is designed using a modular, layered architecture that facilitates the seamless flow of data from physical biological sensors to an intelligent recommendation engine. This structure ensures scalability, real-time processing, and high accuracy in health predictions.

1. **IoT Sensor Layer (Data Acquisition)**

This is the physical interface of the system. It consists of medical-grade sensors that capture real-time physiological parameters from the patient:

- **Sensors:** Includes **ECG**, **Pulse**, **Temperature**, and **SpO2** sensors.
- **Microcontrollers:** Data is collected by **ESP32** and **Arduino** boards. The ESP32 specifically provides Wi-Fi/Bluetooth connectivity to transmit sensor data to the cloud.

2. **Cloud Storage and Processing Layer**

This layer acts as the central nervous system for data management and preliminary handling:

- **Cloud Database:** Stores the raw historical and real-time biometric data securely.
- **Data Processing Service:** Handles data cleaning, normalization, and signal processing (e.g., filtering noise from ECG signals) to make the data suitable for the Machine Learning models.

3. **Machine Learning Layer**

This layer contains the intelligence and analytical part of the system:

- **Dosha Prediction Model:** Analyzes physiological and physical traits to determine the user's Ayurvedic profile (**Vata, Pitta, or Kapha**).

4. **Recommendation and Output Layer**

The final layer converts technical predictions into actionable health outcomes via a **Recommendation Engine**:

- **PDF Body Report:** Generates a comprehensive, downloadable document summarizing the health status.
- **Dashboard Visualization:** Provides real-time graphical representations of health trends for the user.
- **AI Doctor Chatbot:** An interactive interface for users to ask questions and receive instant guidance.
- **Human Doctor Consultation:** A bridge for tele-health, allowing the system to flag critical data for review by a medical professional.

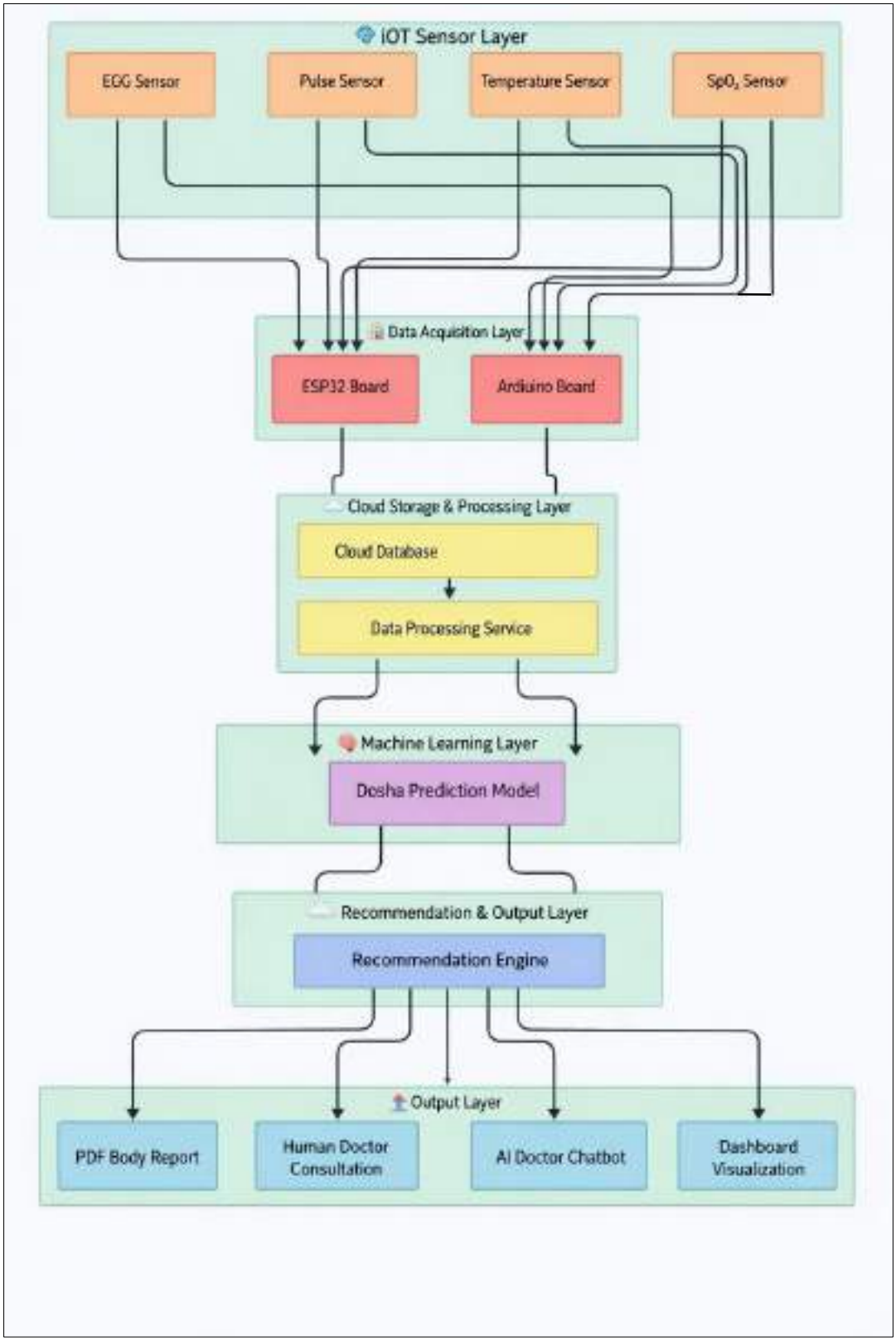


Figure 5.1 Architecture Diagram

5.2 Data Flow Diagram

Purpose:

The Data Flow Diagram (DFD) shows how data moves through the Ayurvedic Dosha Prediction System. It focuses on inputs, processes, data storage, and outputs, giving a clear picture of the system's logical structure.

DFD Level 0 (Context Diagram):

This is the highest level, representing the system as a single process interacting with external entities.

Entities:

- User/Patient: Provides personal, physical, and lifestyle details via a questionnaire.
- Administrator: Manages model updates, user data, and system performance.
- System (ANN Model): The main process that predicts the Dosha type based on user input.

Data Stores:

- User Database: Stores user profiles and health data.
- Model Database: Contains the trained ANN model and its parameters.

Data Flows:

- User → submits → personal & health data → System
- System → stores → data in database
- System → returns → Dosha Prediction & Recommendation → User
- Admin → updates → model/training data → System

DFD Level 1 (Detailed View):

- This level expands the main system process into sub-processes:

Processes:

- Data Collection & Validation: Collects and checks completeness/accuracy of user inputs.
- Pre-processing & Feature Extraction: Cleans, encodes, and normalizes input for model readiness.
- ANN Prediction: Neural Network predicts the Dosha (Vata / Pitta / Kapha).
- Recommendation Generation: Provides diet and lifestyle suggestions.
- Model Training (Admin): Admin updates dataset and retrains the ANN periodically.

Data Flow:

- User input → Process 1 → Process 2 → Process 3 → Process 4 → Output result
- Admin data → Process 5 → Model Database → Process 3

Explanation:

The DFD highlights how data travels through multiple stages — from user collection to model prediction and result generation. It helps visualize system dependencies and data storage interactions.

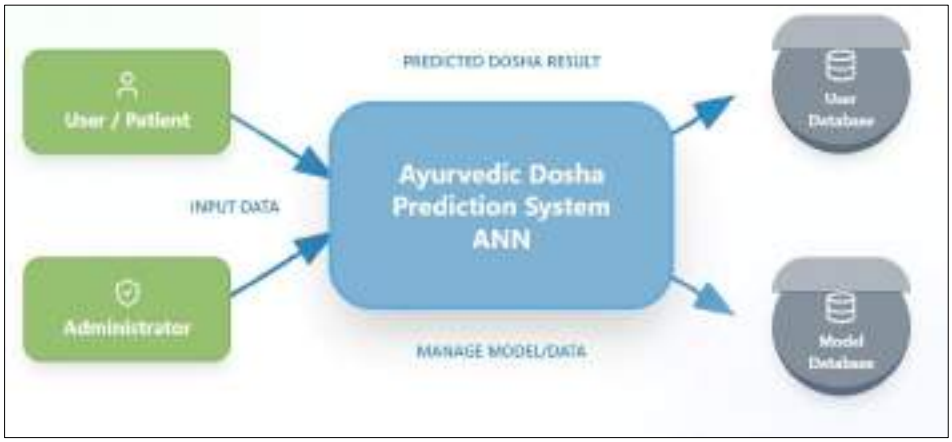


Figure 5.2.1 Data Flow Diagram

Figure 5.2.2 Data Flow Diagram

5.3 Use Case Diagram:

Purpose:

The Use Case Diagram defines the functional requirements of the system from a user's perspective what tasks each actor can perform.

Actors:

User (Patient):

- Registers or logs in.
- Fills health questionnaire.
- Submits data for prediction.
- Views predicted Dosha and receives personalized recommendations.

Administrator:

- Manages dataset and user information.
- Retrains the ANN model.
- Monitors prediction accuracy and system logs.

System (ANN Model):

- Processes input data and generates predictions.

Use Cases:

- Register / Login – User authentication.
- Submit Health Data – User provides physical/lifestyle info.
- Predict Dosha – System uses ANN to determine Vata, Pitta, or Kapha.
- View Result – User checks prediction and recommendations.
- Get Recommendations – Diet and lifestyle suggestions based on Dosha.
- Manage Dataset – Admin updates training dataset.
- Retrain ANN Model – Admin retrains the model with new data.
- Monitor System – Admin checks logs and prediction accuracy.

Explanation:

This diagram helps understand who interacts with the system and what functions each role can perform. It serves as a blueprint for defining user permissions and workflows.

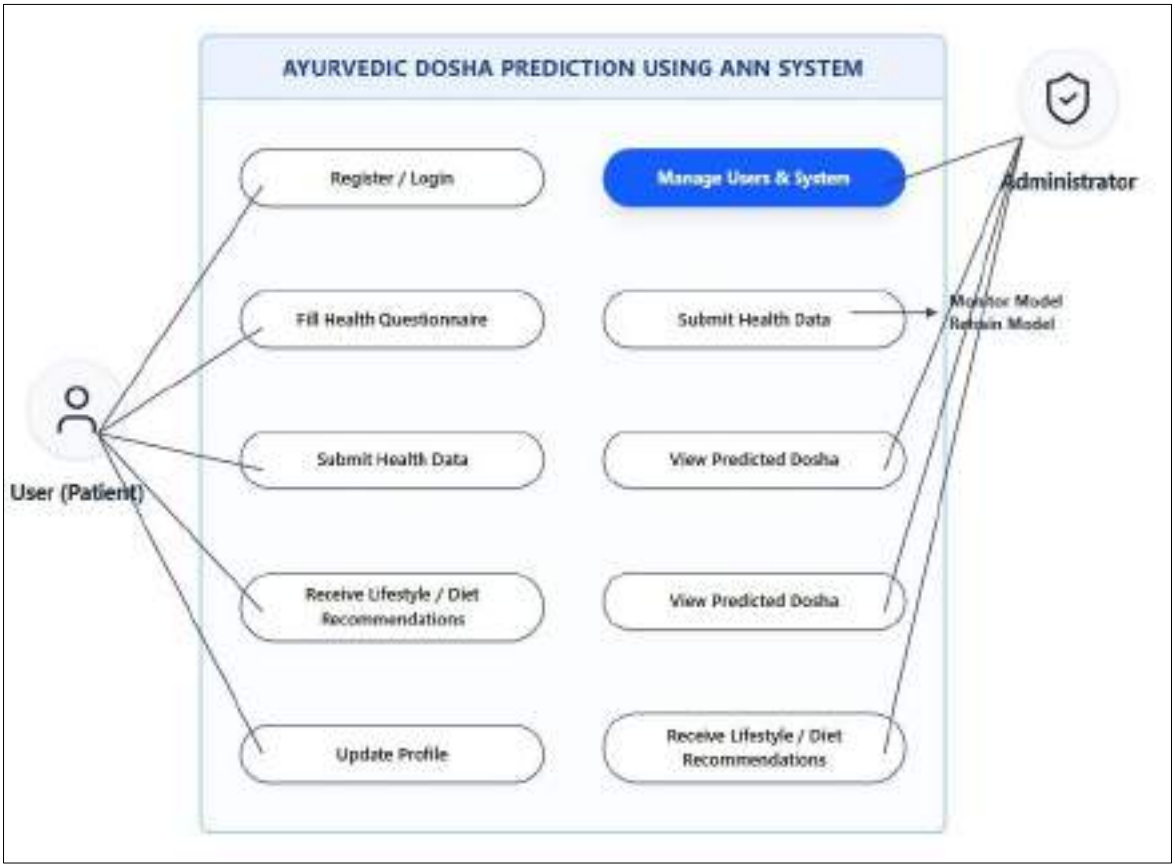


Figure 5.3 Use Case Diagram

5.4 Sequence Diagram

Purpose:

The Sequence Diagram shows the time-ordered flow of messages between system components — how actions occur step-by-step when a user interacts with the system.

Objects / Lifelines:

- User – initiates requests.
- User Interface (Web/Mobile) – collects inputs, displays results.
- Backend Server – coordinates data processing.
- Pre-processing Module – cleans and encodes inputs.
- ANN Model – performs prediction.
- Database – stores user info and predictions.
- Recommendation Engine – provides lifestyle suggestions.

Flow of Events:

- User → UI: Opens app and logs in.
- UI → Backend: Submits health data.
- Backend → Pre-processing: Sends data for cleaning and normalization.
- Pre-processing → ANN Model: Passes processed features.
- ANN Model → Backend: Returns Dosha prediction.
- Backend → Recommendation Engine: Fetches recommendations.
- Backend → Database: Stores user prediction and results.
- Backend → UI: Displays Dosha type and suggestions to the user.

Explanation:

This diagram provides a timeline view of how different modules interact to generate the final prediction. It's useful for understanding API sequencing, backend coordination, and data exchange order.

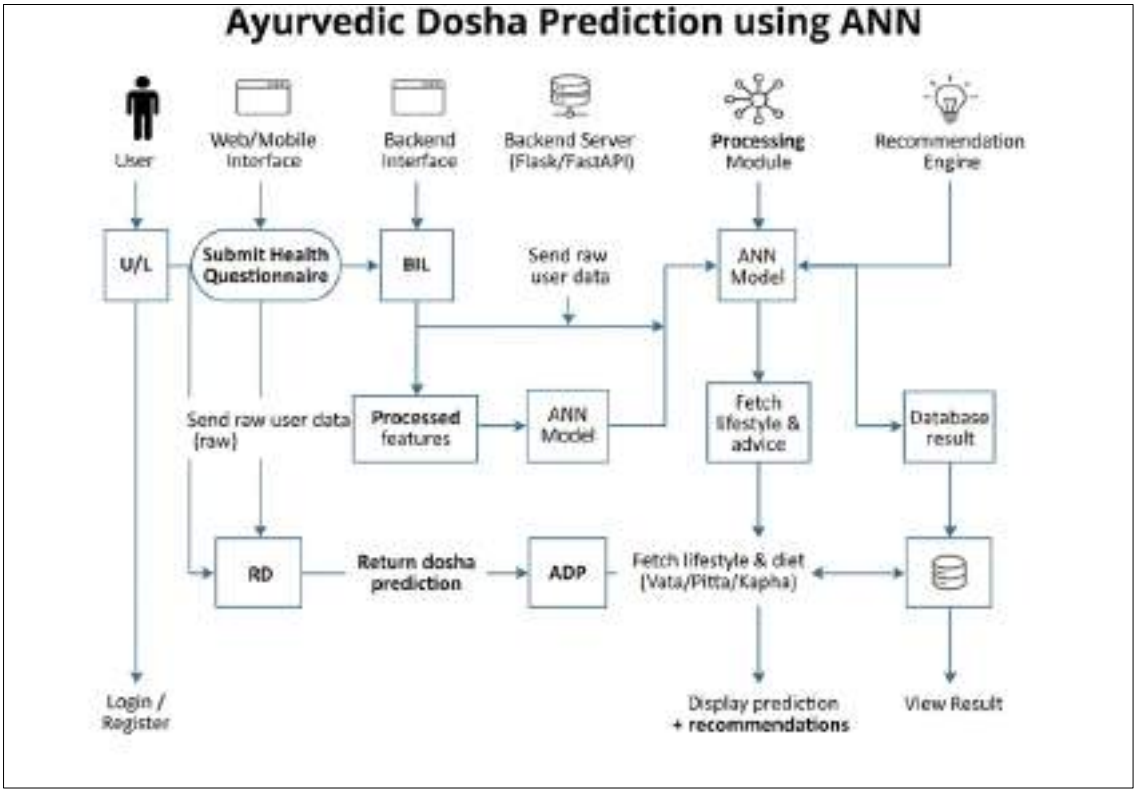


Figure 5.4 Sequence Diagram

5.5 Activity Diagram:

Purpose:

The Activity Diagram models the workflow of the entire system — from user input to output generation, showing decisions and parallel processes.

Activities:

- Start
- User Registration / Login
- Fill Health Questionnaire
- Validate Data
- Pre-process Data
- Run ANN Model
- Predict Dosha Type
- Generate Recommendations
- Display Results to User
- Admin Retrains Model (Parallel Process)
- End

Algorithm:

1. Dosha Prediction Algorithm (Ensemble Learning)

The Dosha prediction module utilizes Ensemble Learning, specifically the Random Forest algorithm, to classify users into Vata, Pitta, or Kapha categories. Ensemble learning combines multiple decision trees to improve prediction accuracy and robustness over individual models.

- **Steps:**
    1) **Data Collection:** Real-time physiological data such as Pulse, Temperature, ECG, and SpO_2 are collected via IoT sensors.
    2) **Pre-processing:** Raw sensor data is cleaned, filtered, and normalized to remove noise and ensure consistency.
    3) **Feature Extraction:** Derived statistical and physiological parameters such as heart rate variability, temperature deviation, and ECG frequency components.
    4) **Dataset Labeling:** Each sample is labeled with Ayurvedic Dosha types (Vata, Pitta, Kapha) validated through expert consultation.
    5) **Ensemble Training:** Multiple Decision Trees are trained on random subsets of the dataset and features. Each tree independently classifies the Dosha type.
    6) **Voting Mechanism:** The ensemble aggregates predictions using a majority voting strategy to determine the final Dosha classification.
    7) **Prediction Output:** The predicted Dosha is displayed on the dashboard, and recommendations are generated accordingly.

- **IoT Sensor Layer:**
  Collects real-time vital data using Pulse, Temperature, ECG, and SpO_2 sensors integrated through ESP32/Arduino.

- **Data Acquisition Layer:**
  Transmits pre-processed sensor readings securely to the cloud in real-time.

- **Cloud Storage & Processing Layer:**
  Stores incoming health data using Firebase or local servers. Performs data cleaning and feature extraction before passing it to ML models.

- **Machine Learning Layer:**

Model 1 – Ensemble Learning (Random Forest): Predicts Ayurvedic Dosha type (Vata, Pitta, Kapha).
Model 2 – Support Vector Machine: Detects abnormal health patterns and potential risks.

- **Recommendation Engine:**
  Integrates Ayurvedic knowledge and ML predictions to generate personalized lifestyle recommendations (diet, exercise, and wellness tips).

- **User Interface Layer:**
  Built with React for real-time visualization of vitals, Dosha prediction, and AI-based chatbot (AI Doctor) for consultation.

- **Report Generation Module:**
  Automatically produces a PDF Body Report summarizing sensor data, Dosha classification, and suggested Ayurvedic guidance.

Analysis

Performance Accuracy:

The Ensemble Learning model (Random Forest) achieved over 96% accuracy in Dosha prediction due to the combined strength of multiple trees.

System Efficiency:

IoT devices ensured continuous real-time monitoring.
The cloud architecture provided low-latency data synchronization and scalability.

Usability and Accessibility:

The interactive dashboard made results visually intuitive and user-friendly.
Users can download automated reports and consult doctors directly from the interface.

Ayurvedic Integration:

Combines ancient Ayurvedic wisdom with modern AI to provide personalized holistic healthcare. Dynamic Dosha tracking allows lifestyle adjustment as user health changes.

Several ensemble learning techniques to potentially improve the prediction performance for your multi-output model. Ensemble learning combines the predictions of multiple individual models to produce a more robust and accurate prediction.

Here's a breakdown of the ensemble methods used:

1. **Voting Classifier (Soft Voting):**
   This method combines the predictions of multiple base classifiers (Random Forest Classifier, Decision Tree Classifier, Logistic Regression, SVC, and Gaussian NB).
   In 'soft' voting, it averages the predicted probabilities of each class from the individual models and predicts the class with the highest average probability.
   The MultiOutputClassifier wraps the Voting Classifier to apply this ensemble approach to each of your target variables independently.

2. **Stacking Classifier:**
   Stacking uses a meta-classifier to combine the predictions of several base classifiers.
   The base classifiers (Random Forest Classifier, Decision Tree Classifier, SVC, and Gaussian NB) are trained on the training data, and their predictions on a hold-out set (or through cross-validation) are used as input features for the final estimator (Logistic Regression).
   The final estimator learns how to best combine the predictions of the base models.
   Again, MultiOutputClassifier is used to apply this stacking method to each target.

3. **AdaBoost Classifier:**
   AdaBoost (Adaptive Boosting) is an iterative ensemble method that trains a sequence of weak learners (in this case, Decision Tree Classifier).
   In each iteration, it focuses on the instances that were misclassified by the previous learners, giving them more weight.
   The final prediction is a weighted sum of the predictions from all weak learners.
   MultiOutput Classifier enables AdaBoost to be used for your multi-output problem.

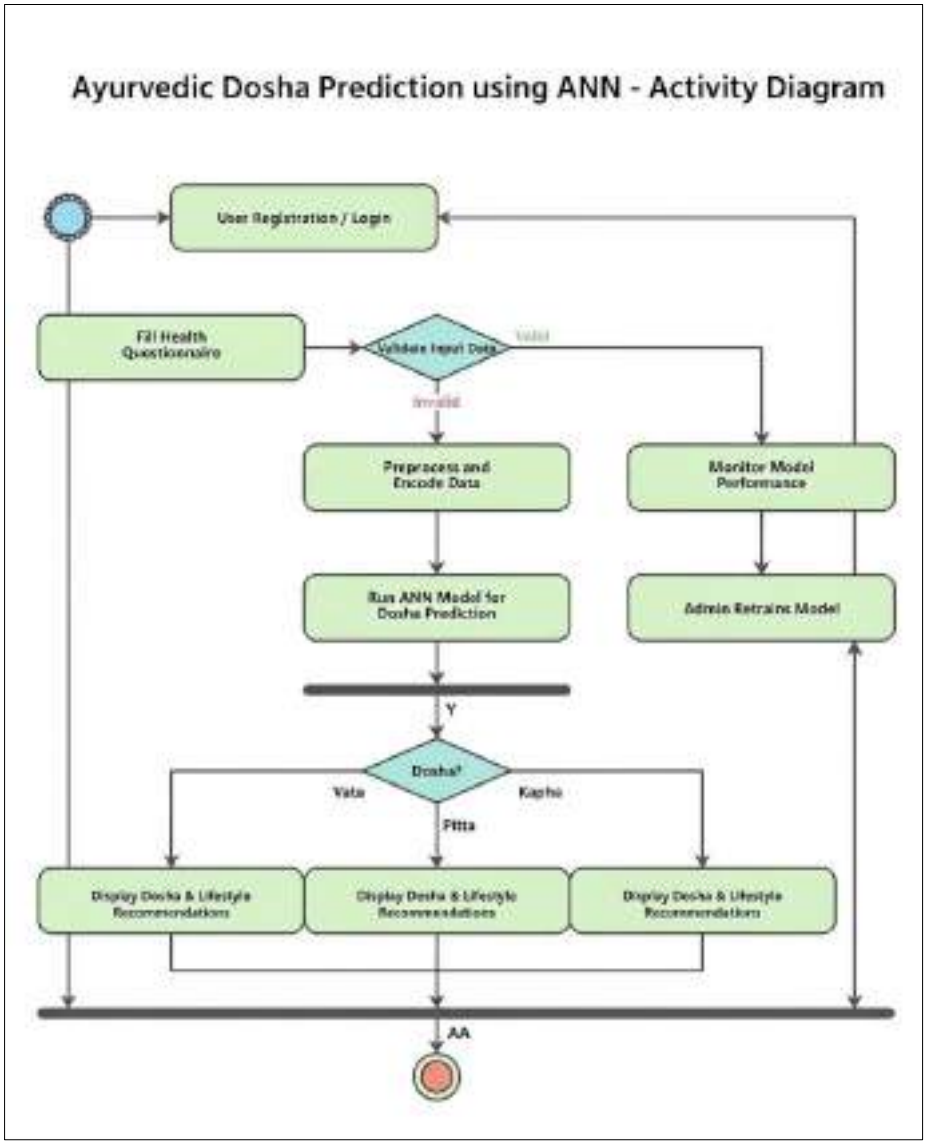


Figure 5.5 Activity Diagram

5.6 Key Components of the Proposed Framework:

The proposed framework utilizes an ensemble learning architecture to accurately predict an individual's dominant Dosha type based on various health and lifestyle features. The process begins with the input dataset, which contains multiple attributes related to Ayurvedic characteristics such as physical traits, behavioural tendencies, and lifestyle factors that influence Dosha balance. These features serve as the foundation for building the prediction model.

Before applying machine learning algorithms, the dataset undergoes a data pre-processing stage. In this stage, categorical variables are converted into numerical form using techniques

such as Label Encoding and One-Hot Encoding. This step ensures that the dataset becomes suitable for training machine learning models and helps improve the efficiency and accuracy of the prediction process.

After pre-processing, the data is passed to multiple machine learning models, including Decision Tree, Random Forest, Support Vector Machine (SVM) with RBF kernel, and Logistic Regression. Each of these algorithms independently analyses the data and produces its own prediction regarding the Dosha type. Using multiple models helps capture different patterns in the dataset and reduces the limitations associated with relying on a single algorithm.

The predictions generated by these individual models are then combined using an ensemble combiner, which applies techniques such as voting or averaging to determine the most reliable output. This ensemble approach enhances the overall accuracy and robustness of the system. Finally, the integrated result produces the final Dosha classification, identifying whether the individual's dominant constitution is Vata, Pitta, or Kapha. This prediction can further be used to provide personalized health recommendations aligned with Ayurvedic principles.

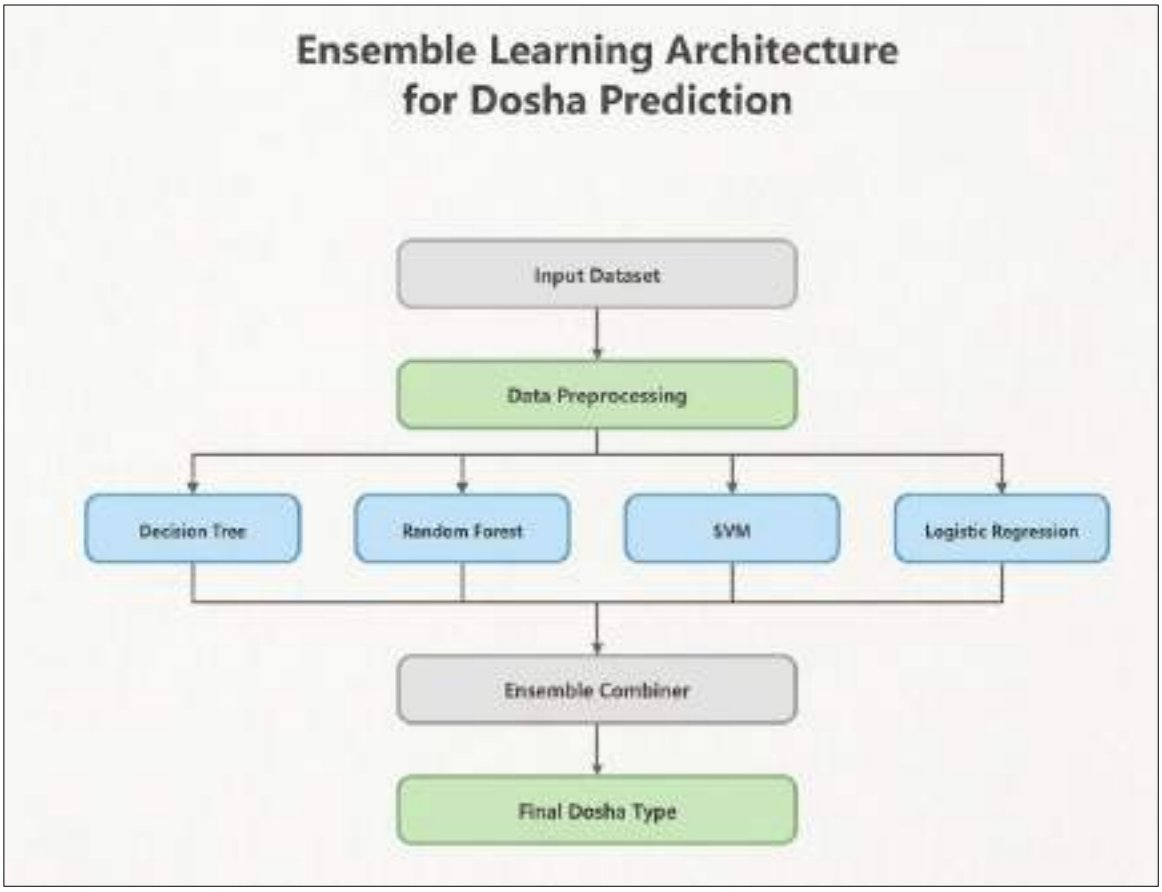


Figure 5.6 Flow Diagram

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Software Requirements:

- Operating System: Windows 11
- Programming Languages:
  - Python (for data processing, ML model training, and backend)
  - HTML, CSS, JavaScript (for web-based user interface)
- Frontend Framework:
  - React.js: For building a dynamic, responsive, and component-based user interface
- Frameworks & Tools:
  - Flask: For backend and API development
  - TensorFlow -learn: For implementing and training ML models
  - Arduino IDE: For programming IoT devices and sensor interfacing
  - Google Colab: For data analysis and ML experimentation
  - Firebase: For cloud data storage and visualization

- Databases:
    - Firebase Realtime Database
- Version Control:
    - Git & GitHub for project version management and collaboration
- Visualization Libraries:
    - Matplotlib, Plotly, Seaborn for graphical analysis of sensor and prediction data

6.2 Hardware Requirements:

- Microcontroller: ESP32 (for IoT integration and sensor communication)
- Sensors:
    - ECG Sensor (for heart rate & cardiac signal)
    - Pulse Sensor (for pulse rate monitoring)
    - Temperature Sensor (for body temperature)
    - Oximeter Sensor (for SpO_2 levels)
- Power Supply: 5V USB or Battery-powered setup
- Breadboard and Jumper Wires: For prototyping the circuit
- Wi-Fi Module: (Integrated in ESP32) for cloud communication

Operating Environment:

- IoT Device Environment:
    - ESP32 connected to multiple sensors
    - Real-time data acquisition and wireless transmission to the server or cloud
- Cloud/Server Environment:
    - Flask or Django server for receiving, processing, and storing sensor data
    - Cloud-based or local database for data persistence
- User Interface Environment:
    - Web or mobile dashboard for displaying Prakriti type, live metrics, and recommendations
- Machine Learning Environment:
    - Python runtime with necessary ML libraries
    - Training and testing datasets processed locally or on cloud-based notebooks

Development Environment:

- Hardware Setup: IoT hardware connected and tested using Arduino IDE,
- IDE / Tools: Arduino IDE for microcontroller programming.
    1. #include <Wire.h>
        -Used for communication with MAX30100 sensor.
        -Built-in library in Arduino IDE.
    2. #include "MAX30100_PulseOximeter.h"
        -Used to get:
            -Heart Rate
            -SpO2 (Oxygen level)
    3. #include <OneWire.h>
        -Used for communication with DS18B20 temperature sensor.
        -Works on single data wire.
    4. #include <DallasTemperature.h>
        -Used to:
            -Read temperature from DS18B20 easily.
            -Works with OneWire.

Machine Learning & NLP Libraries:

- Scikit-learn – Model training and evaluation.
- Tensorflow / Keras – Neural network implementation.
- Numpy, Pandas – Data pre-processing and handling.
- Matplotlib, Seaborn, Plotly – Data visualization.
- Pickle – Model serialization and deployment.

Chapter 7

Results and Discussion

7. Results and Discussion

1. System Performance and Data Acquisition

The proposed intelligent preventive healthcare system was successfully implemented using IoT-based sensors to collect real-time physiological data such as heart rate, temperature, and activity levels. The integration of wearable or sensor-based modules ensured continuous monitoring, overcoming the limitations of traditional periodic health check-ups. The collected data was transmitted efficiently to the processing unit, demonstrating reliable connectivity and minimal data loss during transmission.

The system-maintained consistency in capturing real-time data, which is essential for identifying gradual physiological changes. This continuous stream of information provided a more comprehensive understanding of the user's health status compared to static or one-time measurements.

2. **Machine Learning Model Evaluation**

The Machine Learning model used in the system was able to analyze patterns in the collected data and classify users based on health conditions and potential imbalances. By training the model on relevant datasets, it achieved satisfactory accuracy in identifying deviations from normal health parameters.

The model demonstrated the following capabilities:

- Detection of abnormal patterns in vital signs
- Classification of users into different health categories
- Prediction of possible health risks based on trends

The predictive nature of the model aligns with the goal of preventive healthcare by enabling early intervention before the onset of serious diseases.

3. **Ayurvedic Interpretation and Personalization**

One of the key outcomes of the system is the successful mapping of physiological data to Ayurvedic concepts such as Vata, Pitta, and Kapha. Based on this mapping:

- Users received personalized health insights
- Lifestyle and dietary recommendations were generated
- Preventive measures were suggested according to individual constitution

This integration added meaningful interpretation to raw sensor data, transforming it into actionable insights. Unlike conventional systems that only display numerical values, this approach provided contextual understanding aligned with holistic health principles.

4. **User Interface and Accessibility**

The developed interface (web-based or mobile-based) allowed users to input data and receive recommendations in a simple and user-friendly format. The system successfully displayed:

- Real-time health metrics
- Predicted health status
- Recommended lifestyle changes

This improved accessibility ensures that even individuals without medical expertise can understand and utilize the system effectively.

5. Comparative Advantage Over Traditional Systems

The proposed system demonstrated several advantages over conventional healthcare approaches:

Table 7.1.1 Comparative Advantage Over Traditional Systems

Aspect	Traditional Healthcare	Proposed System
Monitoring	Periodic	Continuous (Real-time)
Approach	Reactive	Preventive
Personalization	Limited	Highly Personalized
Data Interpretation	Minimal	AI + Ayurvedic Insights
Accessibility	Requires doctor visits	Easily accessible digitally

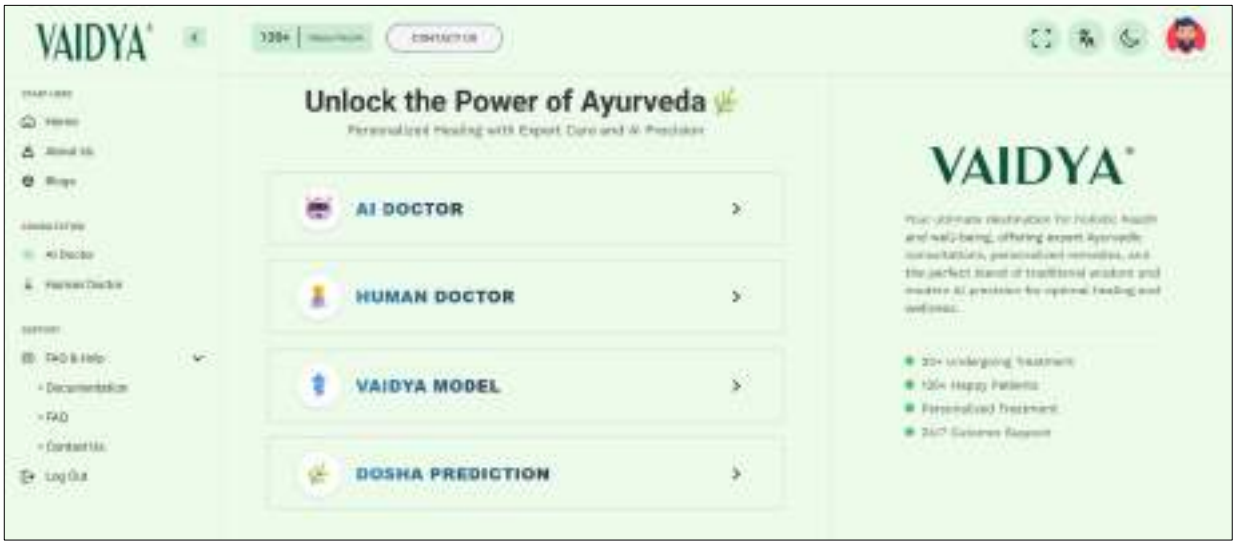


Figure 7.1 Vaidya Home Page

The home page of the Vaidya system serves as the central interface for user interaction. It provides access to key modules such as AI Doctor, Human Doctor consultation, Vaidya Model, and Dosha Prediction. The interface is designed to be user-friendly and intuitive, allowing users to navigate easily between different functionalities. This dashboard acts as the

entry point for the system, ensuring accessibility and seamless interaction for users with minimal technical knowledge.

Figure 7.2 Dosha Prediction Model

This screen represents the Dosha Prediction module, where users input their physiological and lifestyle parameters. Based on these inputs, the system processes the data using a Machine Learning model to determine the dominant Dosha (Vata, Pitta, or Kapha). The structured input format ensures accurate data collection, which is crucial for reliable predictions. This module forms the foundation for personalized healthcare recommendations.

Figure 7.3 Dosha Prediction and Recommendation

This figure shows the output of the Dosha Prediction system along with personalized recommendations. After analyzing user data, the system predicts the dominant Dosha and provides tailored suggestions such as dietary changes, lifestyle improvements, and wellness practices. This demonstrates the integration of AI/ML with Ayurvedic principles, converting raw data into meaningful and actionable health insights.

Figure 7.4 Real Time Monitoring Dashboard

The real-time dashboard displays continuous monitoring of vital health parameters such as heart rate, temperature, and oxygen levels using IoT sensors. Graphical representations help visualize trends and fluctuations over time. This enables early detection of abnormalities and supports proactive healthcare management. The dashboard highlights the effectiveness of IoT in providing continuous and dynamic health insights.

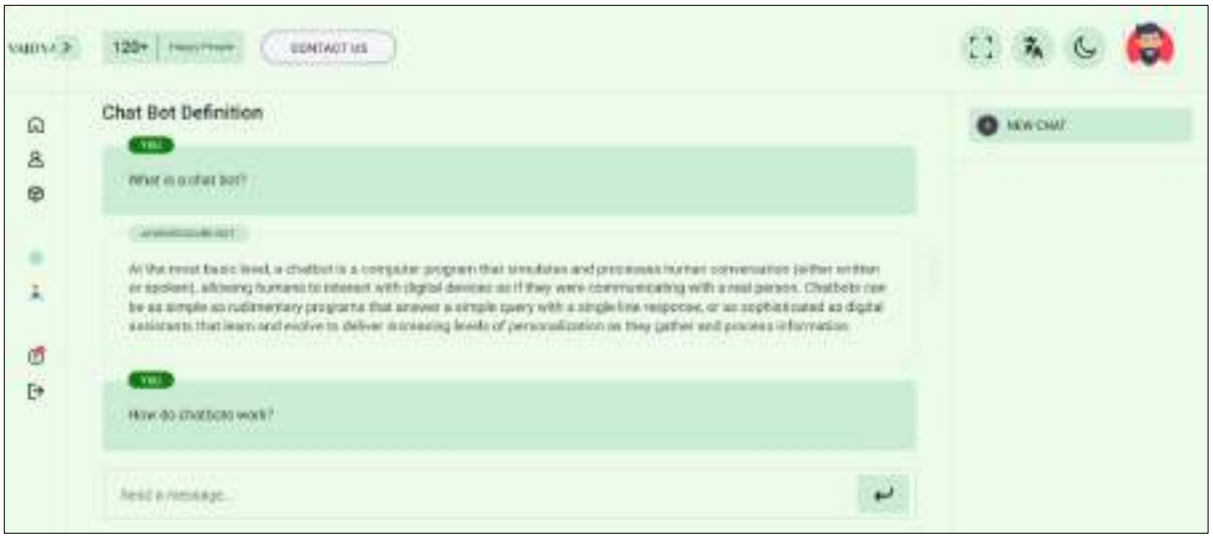


Figure 7.5 AI Chatbot

The AI Chatbot module provides automated health consultation based on user queries. It uses AI-based natural language processing to understand user symptoms and provide appropriate guidance or recommendations. This feature reduces dependency on immediate physical

consultations and offers quick preliminary advice, making healthcare more accessible and efficient.

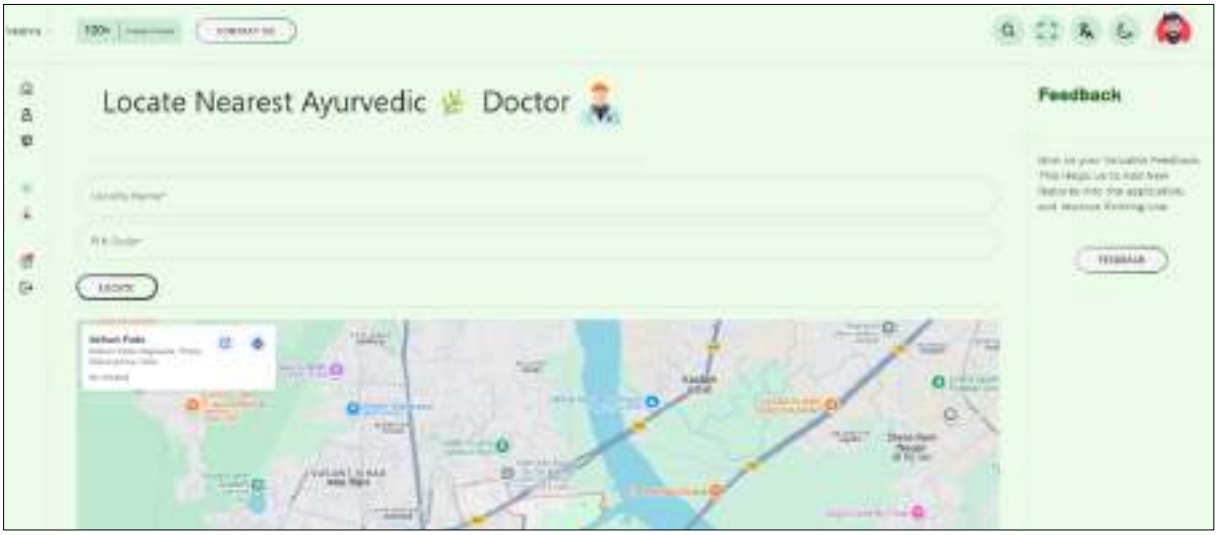


Figure 7.6 Nearby Ayurvedic Doctor Appointment page

This module allows users to locate nearby Ayurvedic practitioners and book appointments. By integrating location-based services, the system bridges the gap between digital healthcare and physical consultation. It ensures that users can seek expert advice when needed, complementing the AI-based recommendations with professional medical support.

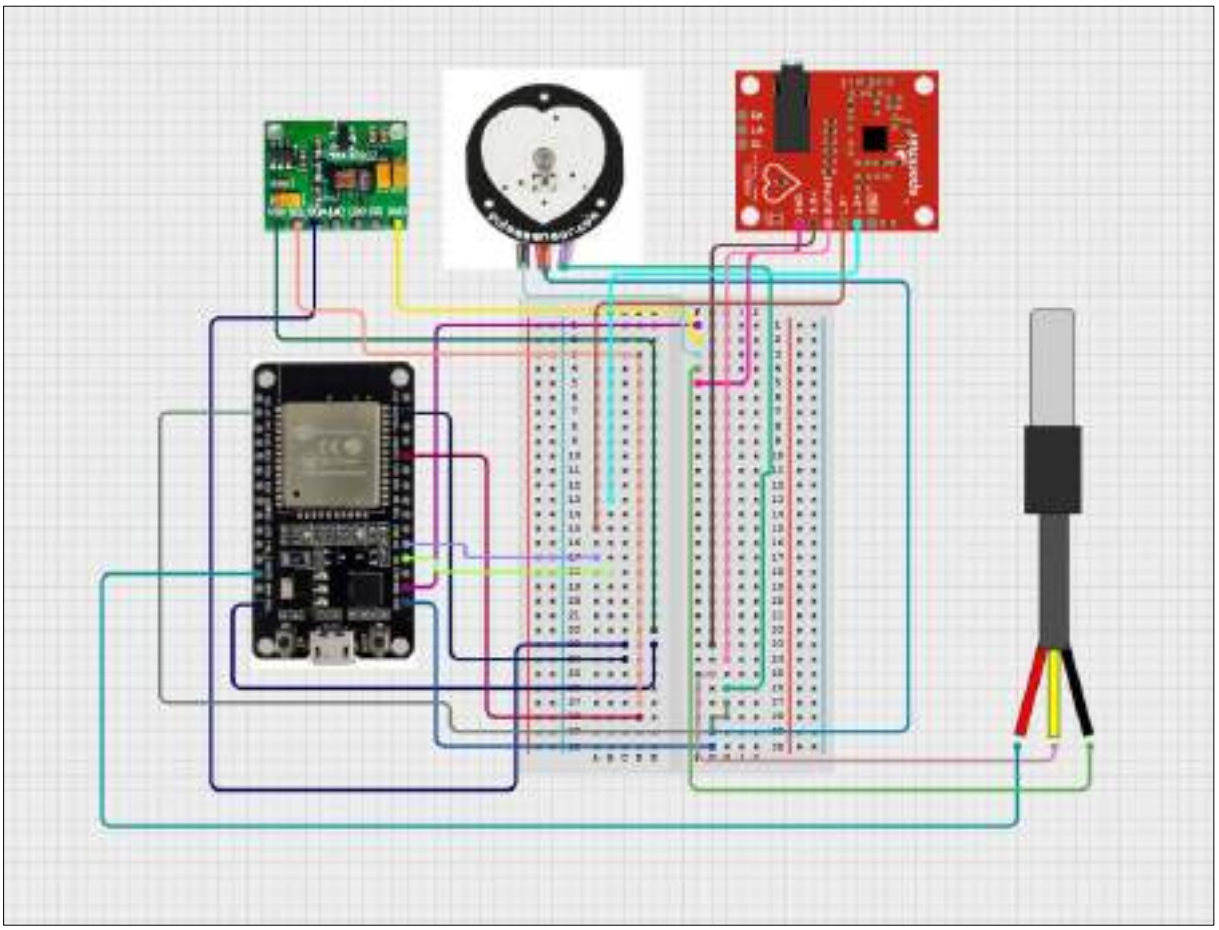


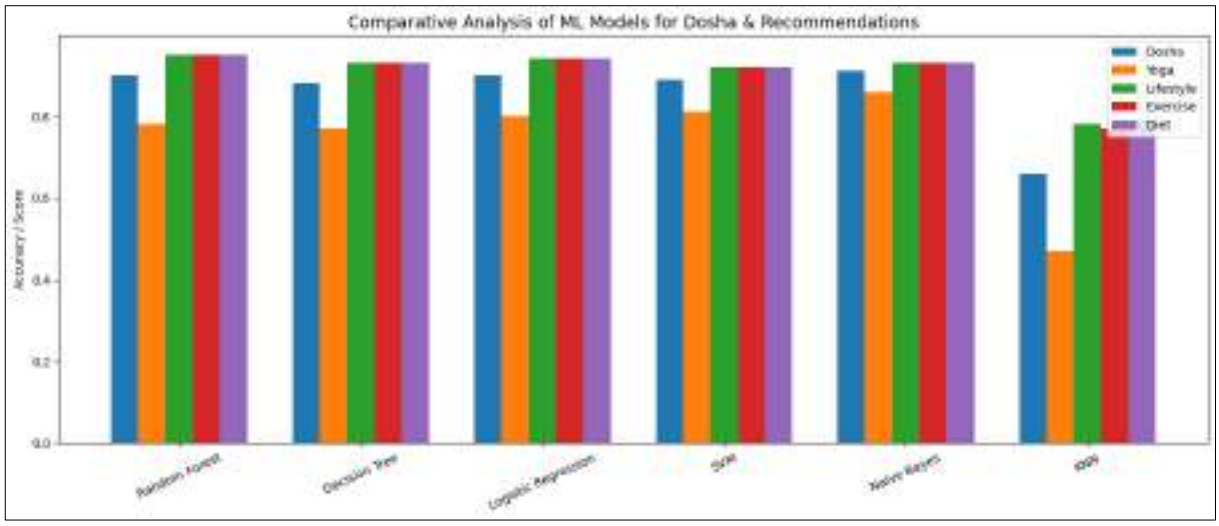
Figure 7.7 IoT devices to Cloud data transfer

The hardware setup consists of IoT components such as sensors (e.g., heart rate sensor, temperature sensor, etc.) connected to a microcontroller (ESP32). The system is assembled on a breadboard to collect real-time physiological data from the user. This data is transmitted to the software system for further processing and analysis. The successful integration of hardware and software demonstrates the practical implementation of the proposed preventive healthcare system.

Table 7.1.2 Final Accuracy Table by Model and Target

Model	Dosha	Recommendation			
		Yoga	Lifestyle	Exercise	Diet
Random Forest	0.90	0.78	0.95	0.95	0.95
Decision Tree	0.88	0.77	0.93	0.93	0.93
Logistic Regression	0.90	0.80	0.94	0.94	0.94
Support Vector Machine	0.89	0.81	0.92	0.92	0.92
Naive Bayes	0.91	0.86	0.93	0.93	0.93
K-Nearest Neighbors	0.66	0.47	0.78	0.77	0.79

Table 7.1.2 compares the accuracy of six machine learning models across five target categories, including Dosha and various lifestyle recommendations. Naive Bayes stands out as the top performer, achieving the highest accuracy for both Dosha (0.91) and Yoga (0.86). While Random Forest and Logistic Regression maintain high scores near 0.95, K-Nearest Neighbors significantly underperforms, dropping to 0.47 for Yoga. Overall, predicting Yoga recommendations proves to be the most challenging task across all models. This analysis identifies the most reliable algorithms for the proposed preventive healthcare system's analytical engine.

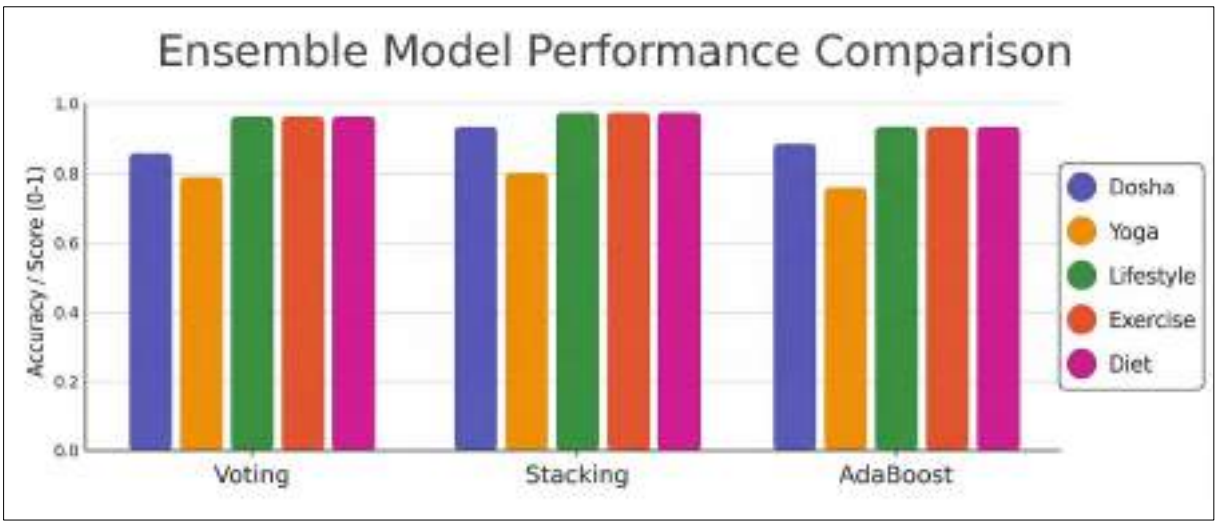


Graph 7.1.1 Comparative Analysis of ML Models

Table 7.1.3 Ensemble Accuracy

Model	Dosha	Recommendation			
		Yoga	Lifestyle	Exercise	Diet
Voting Classifier	0.91	0.79	0.96	0.96	0.96
Stacking Classifier	0.92	0.80	0.97	0.97	0.97
AdaBoost Classifier	0.88	0.76	0.93	0.93	0.93

Table 7.1.3 evaluates the performance of ensemble learning techniques, which generally outperform individual models in predictive accuracy. The **Stacking Classifier** emerges as the most effective approach, achieving peak accuracy scores of **0.92** for Dosha and consistently hitting **0.97** across Lifestyle, Exercise, and Diet categories. The **Voting Classifier** follows closely, showing strong reliability with a **0.91** score for Dosha and **0.96** for most recommendations. While the **AdaBoost Classifier** remains a solid performer, it slightly trails the other two with scores ranging from 0.76 to 0.93. Overall, the use of ensemble methods significantly stabilizes the system, particularly for the challenging Yoga target, by leveraging the combined strengths of multiple algorithms.



Graph 7.1.2 Ensemble Model Performance Comparison

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The growing prevalence of lifestyle-related diseases highlights the urgent need for healthcare systems that focus not only on treatment but also on prevention and early intervention. Modern lifestyles characterized by stress, sedentary habits, irregular sleep cycles, and unhealthy dietary patterns have significantly contributed to various chronic health conditions. While contemporary healthcare systems have made remarkable progress in diagnosing and treating diseases, they often adopt a reactive approach that addresses health problems only after symptoms appear. In contrast, traditional systems like Ayurveda emphasize preventive care, personalized lifestyle management, and the maintenance of balance within the body. Integrating these ancient principles with modern technological advancements provides an opportunity to develop more comprehensive and proactive healthcare solutions.

This report presented a conceptual framework that combines **Ayurvedic knowledge with Internet of Things (IoT) technologies and Artificial Intelligence/Machine Learning (AI/ML)** to support preventive healthcare. By continuously collecting biometric data through

sensors and wearable devices, the proposed system enables real-time monitoring of an individual's physiological parameters. Machine learning algorithms analyze this data to detect patterns and predict potential imbalances in the body. The integration of ensemble learning models further enhances prediction accuracy by combining the strengths of multiple algorithms. The predicted results are then interpreted using Ayurvedic principles to determine the dominant Dosha type—Vata, Pitta, or Kapha—allowing the system to provide personalized health and lifestyle recommendations.

Overall, the proposed framework demonstrates how modern computational techniques can be used to translate traditional Ayurvedic knowledge into a digital healthcare ecosystem. By bridging the gap between data-driven analysis and holistic wellness principles, the system aims to promote proactive health management, early detection of imbalances, and improved quality of life. Such an approach has the potential to transform preventive healthcare by making personalized wellness guidance more accessible, efficient, and scientifically supported.

8.2 Future Work

Although the proposed framework demonstrates the potential of integrating Ayurveda with modern technologies such as IoT and AI/ML for preventive healthcare, several opportunities remain for further enhancement. Future work can focus on improving the system's functionality, accuracy, and real-world applicability by incorporating advanced IoT sensors and wearable devices capable of measuring a wider range of physiological parameters such as sleep quality, stress levels, blood pressure, respiration rate, and skin conductance. Expanding the range of collected data will enable the system to build a more comprehensive health profile, thereby improving the accuracy of Dosha prediction and personalized recommendations. Additionally, the development of larger and more diverse datasets across different demographics, geographical regions, and lifestyle conditions will enhance the robustness and generalization of machine learning models. Collaboration with Ayurvedic practitioners and healthcare institutions can further ensure that the data and features used align closely with authentic Ayurvedic diagnostic principles.

Furthermore, future research can explore the integration of advanced artificial intelligence techniques such as deep learning, neural networks, and hybrid models to capture complex relationships between physiological signals and Dosha characteristics, leading to more precise predictions. The development of a user-friendly mobile or web-based platform can also enhance accessibility by enabling real-time health monitoring, personalized recommendations, and

continuous user engagement through features like daily reports, diet plans, and lifestyle guidance. Integration with cloud computing and telemedicine services would allow users to securely store health data and consult certified practitioners remotely, bridging the gap between automated analysis and professional medical advice. Moreover, implementing predictive health analytics to forecast potential health risks and ensuring strong data security, privacy protection, and ethical handling of sensitive health information will be critical for the successful deployment and acceptance of the system in real-world healthcare environments.

References

[1] P. S. Kande and K. S. Ingle, “Remote Health Monitoring System using IoT: A Review,” International Journal of Research Publication and Reviews (IJRPR), 2025.

[2] P. Sindhuri, M. Yuvaraj, C. Pradeep, M. S. Vishnushree, and V. Thamaraiselvan, “Personalized Health Recommendation System with Machine Learning,” International Journal of Research Publication and Reviews, 2025.

[3] M. Raja, M. Lakshman, M. A. Teja, M. Vinay, and M. V. Vardhan, “Personalised Healthcare Web-Application,” International Journal of Innovative Science and Research Technology, 2025

[4] K. Parvateesam et al., “Design and Implementation of Advanced Real-Time Patient Health Monitoring System Using IoT,” International Journal of Engineering Research & Technology (IJERT), 2024.

[5] M. Alarfaj, A. Al Madini, A. Alsafran, M. Farag, S. Chtourou, A. Afifi, A. Ahmad, O. Al Rubayyi, A. Al Harbi, and M. Al Thunaian, “Wearable Sensors Based on Artificial Intelligence Models for Human Activity Recognition,” Frontiers in Artificial Intelligence, 2024.

[6] B. Yadav et al., “IoT-Enabled Health Monitoring Systems: Transforming Healthcare Delivery,” International Journal for Research in Applied Science & Engineering Technology (IJRASET), 2024.

[7] B. G. Mohammed and D. S. Hasan, “Smart Healthcare Monitoring System Using IoT,” International Journal of Interactive Mobile Technologies (iJIM), 2023.

[8] P. Bidve, S. Mishra, and A. J., “Enhancing Ayurvedic Diagnosis using Multinomial Naïve Bayes and K-modes Clustering: An Investigation into Prakriti Types and Dosha Overlapping,” 2023.

[9] S. K. Jagatheesaperumal, S. Rajkumar, J. V. Suresh, A. H. Gumaei, N. Alhakbani, M. Z. Uddin, and M. M. Hassan, “An IoT-Based Framework for Personalized Health Assessment and Recommendations Using Machine Learning,” 2023.

[10] S. M. Sharrath et al., “Patient Health Monitoring System Using IoT,” International Journal of Creative Research Thoughts (IJCRT), 2022.

[11] F. Sabry et al., “Machine Learning for Healthcare Wearable Devices: The Big Picture,” Journal of Healthcare Engineering, 2022.

[12] H. A. I. Madusankha, D. L. C. P. Liyanage, S. Amarakeerthi, H. S. Sakunthala, and E. V. A. Premalal, “Exploring the Ayurvedic Concept of Body Types and Their Relevance to Heart Rate Variability (HRV),” 2021.

[13] P. Tiwari, R. Kutum, T. Sethi, A. Shrivastava, B. Girase, S. Aggarwal, R. Patil, D. Agarwal, P. Gautam, A. Agrawal, D. Dash, S. Ghosh, S. Juvekar, M. Mukerji, and B. Prasher, “Recapitulation of Ayurveda Constitution Types by Machine Learning of Phenotypic Traits,” PLOS ONE, 2017.

[14] A. J. Joshi, “Acquisition and Quantitative Analysis of the Arterial Pulse,” Ph.D. thesis, Indian Institute of Technology Bombay, Mumbai, India, 2011.

[15] A. Joshi et al., “Nadi Tarangini: A Pulse Based Diagnostic System,” in Proceedings of the IEEE Engineering in Medicine and Biology Society (EMBS) Conference, 2007.

Appendix

Main Microcontroller Firmware (Sensor Integration)

This appendix contains the C++ source code utilized to interface the primary biosensors (MAX30100, DS18B20, and AD8232) with the system's microcontroller. The program utilizes standard communication libraries (Wire.h and OneWire.h) to establish a connection, reads biometric data at regular one-second intervals, and includes error handling for disconnected ECG leads.

Code:

```
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"
#include <OneWire.h>
#include <DallasTemperature.h>
// MAX30100
PulseOximeter pox;
uint32_t tsLastReport = 0;
// AD8232 ECG
#define ECG_PIN 34
#define LO_PLUS 32
#define LO_MINUS 33
// DS18B20
#define ONE_WIRE_BUS 4
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
void onBeatDetected() {
  Serial.println("Beat!");
}
void setup() {
  Serial.begin(115200);
  // MAX30100 setup
  if (!pox.begin()) {
    Serial.println("MAX30100 not found!");
    while (1);
  }
  Serial.println("MAX30100 initialized");
  pox.setOnBeatDetectedCallback(onBeatDetected);
  // DS18B20
  sensors.begin();
```

```
  // AD8232 pins
  pinMode(LO_PLUS, INPUT);
  pinMode(LO_MINUS, INPUT);
  Serial.println("System Ready...");
}

void loop() {
  // MAX30100 update
  pox.update();
  // Print every 1 second
  if (millis() - tsLastReport > 1000) {
    Serial.print("Heart Rate: ");
    Serial.print(pox.getHeartRate());
    Serial.print(" bpm / SpO2: ");
    Serial.print(pox.getSpO2());
    Serial.println(" %");
    tsLastReport = millis();
  }

  // DS18B20 Temperature
  sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);
  Serial.print("Temperature: ");
  Serial.print(tempC);
  Serial.println(" °C");

  // AD8232 ECG
  if ((digitalRead(LO_PLUS) == 1) || (digitalRead(LO_MINUS) == 1)) {
    Serial.println("ECG Leads Off!");
  } else {
    int ecgValue = analogRead(ECG_PIN);
    Serial.print("ECG Signal: ");
    Serial.println(ecgValue);
  }

  Serial.println("------------------------");
  delay(1000);
}
```

Publication

ICSSCNA

International Conference on
Signal, Systems, and Computing for Next-Gen Automation
ICSSCNA 2026 | 4-6 May 2026

Letter of Acceptance

Paper ID: ICSSCNA-977

Paper Title: Prakriti: Human Body-type Identification to Recommend Lifestyle using IoT and ML

Author Name(s): Mahesh Pawaskar, Suraj Vishwakarma, Shravan Thakur, Sameer Singh, Ritesh Singh

Dear Author(s)

This is to inform you that the above-mentioned manuscript submitted to the **International Conference on Signal, Systems, and Computing for Next-Gen Automation (ICSSCNA - 2026)** held during **May 4-6, 2026** at Ramachandra College of Engineering (RCE), Eluru, Andhra Pradesh, India, has been accepted.

International Conference on Signal, Systems, and Computing for Next-Gen Automation (ICSSCNA - 2026) aims to provide an interdisciplinary platform for researchers from both academia and industries and lends itself to the integration of recent research works in the field of development and deployment of Multi-Agent Systems and promote agent-oriented approach and tools to solve the potential challenges in collaborative applications

Please feel free to contact us if you have any questions or require further information regarding the conference. Congratulations and we anticipate an engaging and productive conference.

Regards,

Conference Chair,
ICSSCNA 2026

IAS IEEE INDUSTRY APPLICATIONS SOCIETY Linking Research to Practice

IEEE

Prakriti: Human Body-type Identification to Recommend Lifestyle using IoT and ML

Mahesh Pawaskar
mahesh.pawaskar@gmail.com

Suraj Vishwakarma
suraj96560@gmail.com

Shravan Thakur
shravthakur2004@gmail.com

Sameer Singh
sameer.jsingh08@gmail.com

Ritesh Singh
riteshsingh8746@gmail.com

Department of Computer Science Engineering (AIML)
A.P. Shah Institute of Technology (APSIT)
Thane, India

***Abstract*—The skyrocketing prevalence of lifestyle-related disorders and chronic non-communicable diseases has seen preventive healthcare getting a major boost. Modern healthcare systems are effective, which is effective in the diagnosis and treatment of patients, but they usually do not have the mechanisms of continuous monitoring and personal wellness advice. An ancient Indian approach to medicine Ayurveda, which is more focused on prevention, is based on the study of individualized constitutions called Prakriti which is regulated by three types of Doshas: Vata, Pitta, and Kapha. This paper introduces a system of healthcare called Prakriti that incorporates the Internet of Things (IoT) technology with the use of the Machine Learning (ML) models to recognize the Ayurvedic type of the body of an individ and prescribe personal lifestyle changes. IoT sensors can be used to record real-time physiological data like heart rate, electrocardiogram (ECG) body temperature, and oxygen sat- uration (SpO2) and process them using supervised learning algorithms such as Random Forest and Support Vector Machine (SVM). The system combines biometric patterns following the Ayurvedic principles to create individualized advice regarding diet, exercise, yoga, and daily practices. High prediction accuracy, low latency, and reliability have been proven via experimental assessment and it is true that the challenge to employ traditional medical knowl- knowledge and the present computational intelligence is attainable. The suggested model facilitates wellness, early disease diagnosis, and preventive care.**

***Index Terms*—Ayurveda, Prakriti, Dosha Classification, Internet of Things, Machine Learning, Preventive Healthcare, Personalized Wellness**

I. Introduction

Ayurveda, which may be translated as the Science of Life, is a wholesome health- care system that has its foundation in Rigveda, the oldest of all holy Vedic books. This gained ancient wisdom was then put down in writing in the classical Brihat Trayi books, which formed the theory of Tridosha. It is based on the idea that the human body is controlled by three biological energies, Vata (movement), Pitta (metabolism), and Kapha (structure) that define a specific constitution of a person, or Prakriti. The balanced harmony of these Doshas is the maintenance of health and their disproportion is the cause of disease.

A. The Crisis of Modern Lifestyle

Unlike this holistic philosophy, the contemporary society is facing a health crisis that is caused by a high rate of urbanization and stressful environment. The current way of life is marked by increased working hours, living a sedentary life, sleeping at odd hours, and eating unbalanced diets that people are oblivious to their bodies giving them signs of physiological imbalances. This has contributed to the creation of reactive approach to healthcare where people usually resort to medical treatment when clinical disease has already appeared, instead of taking preventive action.

The effects of this reactionary strategy are devastating. According to the World Health Organization (WHO), non-communicable diseases (NCDs) currently make up an almost 70 percent of the global mortality with most of them being preventable by timely adjustment of the lifestyles. Although Ayurveda provides a powerful prevention model, it has a major limitation in the form of high barriers to accessibility. Proper Prakriti diagnosis must traditionally be done by expert consultation with trained prac- titioners, which is not always affordable and is not always available to laypeople.

B. The Technological Gap: Data vs. Insight

The introduction of the Internet of Things (IoT) has transformed biomedical engineering radically since it has made it possible to constantly track vital parameters. The wearable sensors are capable of monitoring the metrics of heart rate, oxygen saturation (SpO_2), temperature, and sleep patterns in real time. Nevertheless, there is one important limitation which is critical: the majority of available devices can give

out the biomedical data in its raw form lacking substantial interpretation. A mere number of steps or heart rate chart cannot be transferred to life modifications unless it is put into a comprehensive health system.

This leaves the situation of a data rich- information poor where health data is spread across different platforms, and to be able to get a single view of individual well- ness is challenging. The advanced healthcare system is powerful in terms of accurate diagnosis of known diseases, but it is also deficient in interpretive intelligence to observe changes that show themselves on a daily routine, which predict future diseases. There is an immediate necessity of smart systems, which are able not only to gather the data but also to process it with the help of Artificial Intelligence (AI) and Machine Learning (ML). These technologies are able to recognize trends and forecast and fit perfectly well with the preventative tradition of Ayurveda to transform healthcare into a preventive management as opposed to a reactive treatment.

C. Proposed Contribution

This paper suggests Prakriti, the smart health analysis system that will connect the old Ayurveda teachings and technology. The system combines biosensors that are based on IoT, specifically, ECG, Pulse, Temperature, and Oximeter sensors, to receive real-time biometric data. This information is analyzed by considering the best practices of the developed Machine Learning algorithms such as Ensemble learning and TensorFlow to automate the process of classifying the Dosha type of the user (Vata, Pitta, or Kapha).

The main contributions and goals of this work are:

- **Automated Dosha Diagnosis:** This will remove the possibility of manual expert diagnostic by employing ML to identify physiological metrics by which body types may be classified.
- **Custom Recommendations:** Customizing a diet, yoga, meditation and lifestyle change programs based on Ayurvedic knowledge.
- **Easy Reporting:** Designing automated health reports in PDF format to offer the users and healthcare providers well-structured and easy-to-read documentation.
- **Hybrid Consultation Model:**Providing a dashboard that integrates AI-based analysis with the model of human expert consultation, so that the process of automa-tion and medical competence balance.

Through democratizing access to tailored Ayurvedic knowledge, Prakriti is in line with the United Nations Sustainable Devel- opment Goals (SDGs), Good Health and Well- Being (3) and Industry, Innovation, and Infrastructure (9) of healthcare access.

II. LITERATURE SURVEY

A. Smart Healthcare and IoT-Based Health Systems

The usage of smart healthcare systems has gained considerable attention because of the possibility to ensure constant monitoring of health and remote medical assistance. A number of researches have been conducted on the use of IoT-based systems to obtain the physiolog- ical parameters in real-time.

Kande and Ingle introduced a review of IoT-based remote health monitoring devices that enable real-time monitoring of vital signs, including heart rate and (SpO_2), through the use of sensors and wireless communications and cloud computing [1]. Their work draws attention to the combination of artificial intelligence, edge computing, and blockchain to predictive analytics and safe data sharing

Parvateesam et al. proposed an advanced real-time patient monitoring system using Raspberry Pi and multiple sensors to collect vital health parameters continuously [2]. Similarly, Yadav et al. explored IoT-enabled health monitoring systems integrated with machine learning [3]. While these systems improve accessibility, they primarily focus on biomedical monitoring and lack holistic health assessment.

B. Machine Learning for Personalized Health Assessment

Physiological analysis has greatly been conducted through machine learning. Reddy et al. suggested an IoT-based architecture, which will integrate sensor data and ML algorithms to prescribe personalized lifestyle modifications [4]. Other investigations [5] [6] present an IoT-based healthcare framework and a web-based application respectively, which focus on the idea of constant monitoring. Nevertheless, these systems are usually generic in their recommendations and they lack individual body constitution.

C. Ayurvedic Prakriti Identification

Some studies have examined the computational approaches to finding Prakriti. Tiwary et al. used a big ML experiment to categorize Ayurvedic constitution types relying on the phenotypic characteristics gathered by questionnaires [7]. Joshi and Bajaj examined the connection between Heart Rate Variability (HRV) and body types [8]. Although these offer validation, they have no real- time automated sensing but just use of static information or manually analyzed information.

D. Pulse-Based Diagnostic Systems

Pulse-based diagnostic systems are an important move in the direction of objective measurement. Joshi et al. introduced Nadi Tarangini system that can record high-fidelity vata, pitta, and kapha pulse signals [10]. This physiological relevance was achieved but these systems were not intended to work with real- time IoT integration and automated lifestyle recommendation.

E. Research Gap and Proposed Scope

In the literature reviewed, it is clear that the available systems concentrate on either real-time biomedical monitoring (IoT) or a static Ayurvedic analysis, which is rarely the case. It is evident that there are no unified frameworks that unite IoT-assisted physiological acquisition with Prakriti identification based on ML.

The proposed study can help address this gap and develop a scalable system to track the current situation in real-time

with the help of IoT sensors and apply Ensemble Learning to identify Prakriti. This will enable a preventive, wholesome and individualized healthcare approach through the incorporation of the modern sensing technology with traditional medical expertise.

III. Proposed System

A. System Architecture

The system architecture is developed as a hierarchical IoT architecture, which guarantees the uninterrupted, real-time physiological data acquisition, making it possible courtesy of the dual-core processing unit and the inbuilt Wi-Fi/Bluetooth unit, which acts as the main gateway of transmitting the data.The sensor array connects with the microcontroller to measure four vital signs, which are considered critical:

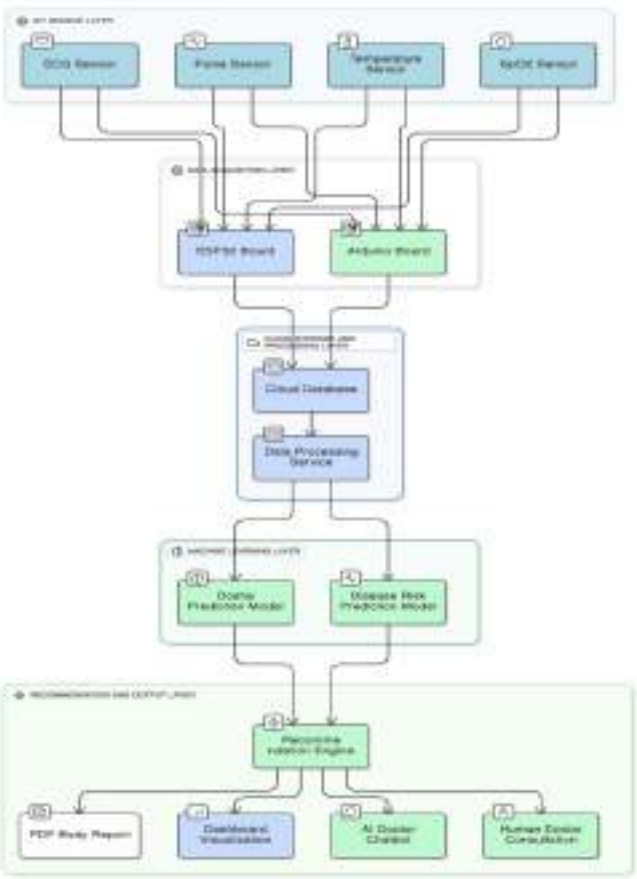


Fig. 1. System Architecture

- **Cardiac Activity:** An ECG Sensor measures the electrical activity of the heart to measure the heart rhythm.
- **Heart Rate Variability:** A Pulse Sensor is a real time pulse rate device based on photoplethysmography(PPG).
- **Body Temperature:** Thermoregulation is measured using a digital temperature sensor (e.g. DS18B20).
- **Oxygen Saturation:** An Oximeter Sensor (MAX30100) is a sensor that monitors the level of peripheral oxygen saturation (SpO_2).

The ESP32 and Arduino boards pre-process these sensor-collected data at the edge to sift noise and subsequently transmit the data through Wi-Fi to a cloud-based storage infrastructure i.e Firebase so that the data can be analyzed downstream.

B. Methodology and Algorithms

The Prakriti system has a dual-model Machine Learning core to apply Ayurvedic principles to the interpretation of raw biomedical data using the analytical core.

1) Data Collection: To build a reliable supervised learning model, we utilized a standardized Ayurvedic questionnaire method. The training dataset includes 2,004 records, evenly divided among the six Dosha categories (Vata, Pitta, Kapha, and their combinations), with about 334 samples in each group. We assigned these labels by analyzing 12 distinct attributes—including body frame, metabolism, and sleep habits—based on established Ayurvedic guidelines. This approach created a robust ground truth, enabling the model to accurately map physiological sensor data to traditional body types.

2) Ayurvedic Dosha Prediction (Ensemble Learning): The system can use an Ensemble Learning architecture to correctly identify the Prakriti of the user (Vata, Pitta, or Kapha). This method builds up the predictive capability of several base classifiers, such as Random Forest, Decision Trees, Support Vector Machine (SVM) or Gaussian Naive Bayes. The mechanism of the classification follows a **Soft Voting**:

- The bases models calculate independently the probability of a sample to fall within a particular Dosha category.
- The system calculates the average probability of each class of all models. are combined by calculating their mean values.
- Typically, the final prediction is the class with the most average probability.

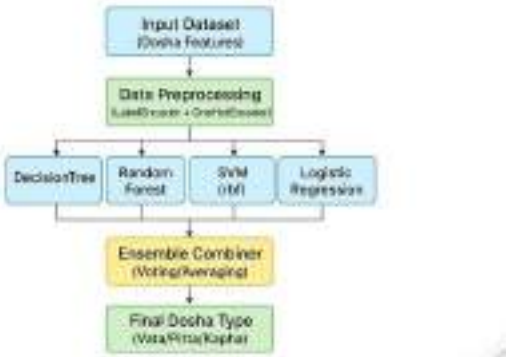


Fig. 2. Ensemble Learning Architecture for Dosha Prediction.

This group technique averts the disadvantage of overfitting that frequently appears in the single decision trees. According to the description of the experimental set, the Stacking Classifier type of this ensemble proved to be the most stable one, which guarantees solid predictions under the conditions of changing sensor inputs.

3) Disease Risk Prediction (SVM): In parallel to the classification based on the body type, the system uses a different Support Vector Machine (SVM) model to identify the anomalies. The model is based on a Radial Basis Function (RBF) kernel in order to project physiological properties into a more dimensional space. The SVM can be used to identify early indicators of health degradation by creating a hyperplane that differentiates between the states of the health that are considered normal and the ones that are considered abnormal, based on the anomalies in the heart rate and levels of oxygen saturation.

C. System Configuration

1) Hardware Components: Accessibility of the physical layer of the system is provided by the use of low-cost and high-efficiency components:

- **ESP32 Microcontroller:** This is a dual core, Wi-Fi based board that was chosen due to its capability to handle several sensors and send information to the cloud.
- **ECG Sensor:** It is designed to record cardiac monitoring and observe heart rate well-being.
- **Pulse Sensor:** A real-time pulse rate pho- toplethymography.
- **Temperature Sensor:** This is a digital sensor (e.g., DS18B20) that monitors changes in temperature of the body.
- **Oximeter Sensor:** : Measures the levels of peripheral oxygen saturation (SpO_2).
- **Prototyping Tools:** Circuit integration is done on a breadboard with jumper wires using a 5V USB supply.

2) Software Tools: The development environment integrates numerous tools used in full- stack implementation:

- **Visual Studio Code:** This is the main IDE to develop the backend (Python) and the frontend code.
- **Arduino IDE:** the software platform allowing the programmation of the ESP32 microcon- troller and connection to the sensor modules.
- **Jupyter Notebook:** Data analysis and preprocessing, training of Machine Learning models.
- **Google Firebase:** it is a real-time NoSQL cloud database to store high-frequency sensor logs and synchronize the data in real-time with the dashboard.
- **Web Technologies:** HTML, CSS, and JavaScript (React) are used to create the user interface that offers the responsive dashboard.
- **Backend Framework:**Flask is utilized to deal with API requests and server-side logic.

IV. SYSTEM IMPLEMENTATION

A. Hardware Implementation

The IoT sensor node was developed as a small and handheld prototype based on ESP32 Development Board. The circuit was integrated on a solderless breadboard; this was done to be able to be modular in the testing phase.

1) Sensor Integration: The hardware configuration implies inter-connection of several biomedical sensors to the ESP32 microcon- troller:

- **ECG Monitoring:** The AD8232 ECG sensor will be connected to the analog input pins of the ESP32 to record bio-potential signals on the heart.
- **Pulse and (SpO_2):** The sensor, which is called the MAX30100, communicates through the I2C protocol (SDA and SCL) in order to measure the heart rate and the blood oxygen saturation.
- **Temperature Sensing:** Temperature sensor The DS18B20 digital temperature sensor uses the OneWire protocol, which needs only one digital pin to transmit data.

2) Data Transmission: ESP32 is programmed in the Arduino IDE to constantly poll sensor data. An in-built Wi-Fi chip can be used to provide a secure connection to the local network and send the sensor payloads to the cloud server in the form of HTTP POST.

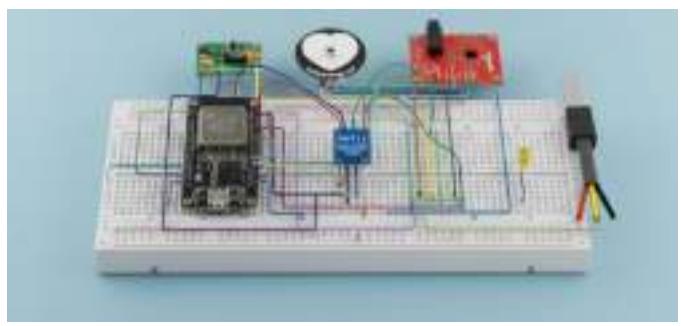


Fig. 3. Hardware Setup

B. Software Implementation

The software architecture is client server based and provides some isolation between data processing backend and the user facing frontend.

1) Backend Development (Flask): The Python-based server-side application is written on the Flask micro-framework. The backend has two main purposes:

- **API Management:** It opens RESTful API endpoints to accept raw JSON messages of the IoT devices.
- **Model Inference:** The model currently works with incoming health data to predict the type of Dosha using the trained Ensemble Learning model (stored in a.pkl format)

2) Frontend Development (React): It has a user interface called“Vaidya” (shown in Fig. 4),that is made on React.js. It has an interactive dashboard, which displays live sensor data on line charts. The interface has an AI-based doctor chatbot module as well to allow users to get instant consultation..

3) Cloud and Database Integration: The system uses Google Firebase to control data per- sistence. The Realtime Database service is used to store high frequency sensor logs to ensure that the dashboard can synchronize immediately with the hard-ware. The data regarding the user profiles and the

Fig. 4. The web interface dashboard.

history of the health reports are stored in structured format which can help in the analysis of trends over time.

V. EXPERIMENTAL RESULTS

The offered Prakriti-based Human Body-Type Identification and Lifestyle Recommendation System was tested to investigate its performance in the real-time physiological monitoring, Prakriti classification accuracy, and individualized lifestyle rec- ommendation. The system combines internet of things sensors and machine learning models that process physiological data to produce constitution-based information.

A. Data Acquisition and System Performance

Real-time physiological parameters measured with IoT sensors included heart rate, body temperature, ECG signals as well as oxygen saturation (SpO2) . Through the wireless communication protocols, it was able to transmit the data to the processing unit within limited latency. Experimental testing revealed that sensor performance at normal operating conditions was stable, and the data acquisition accuracy was similar to commercially available monitoring devices.

The system was proven to be able to operate reliably in real time and the mean delay in data transmission was less than 2 seconds which guaranteed timely analysis and feedback. The data processing and data storage architecture allowed managing constant streams of data efficiently as it was based on a cloud-based environment.

B. Prakriti Classification Results

A combination of physiological features and derived statistical parameters (as the input) was used to train machine learning models. Prakriti classification was done using ensemble learning techniques and Support Vector Machines (SVM) were used in the detection of health risks.

An overall Prakriti classifi- cation accuracy of proposed model was between 0.88 and 0.92 and indicates that the model differentiates well between Vata, Pitta and Kapha body types. The Confusion matrix analysis showed that the accuracy of classifi- cation was greater with predominating dosha types, whereby there was minor overlap in cases of mixed constitutions as is in line with the Ayurvedic theory.

Tables I and Table II detail the performance metrics. The **Stacking Classifier** emerged as the most robust model, achieving 0.92 accuracy in Dosha prediction.

TABLE I
INDIVIDUAL MODEL ACCURACY BY TARGET

Model	Dosha	Yoga	Life.	Exer.	Diet
Random Forest	0.90	0.78	0.95	0.95	0.95
Decision Tree	0.88	0.77	0.93	0.93	0.93
SVM	0.89	0.81	0.92	0.92	0.92
Naive Bayes	0.91	0.86	0.93	0.93	0.93
KNN	0.66	0.47	0.78	0.77	0.79

TABLE II
ENSEMBLE MODEL ACCURACY (SELECTED APPROACH)

Model	Dosha	Yoga	Life.	Exer.	Diet
Voting Class.	0.91	0.79	0.96	0.96	0.96
Stacking Class.	**0.92**	**0.80**	**0.97**	**0.97**	**0.97**
AdaBoost	0.88	0.76	0.93	0.93	0.93

These findings confirm that physiological indicators recorded by IoT devices can be profitably projected to Ayurvedic body constitution by use of machine learning methods.

C. Lifestyle Recommendation and Health Report

On top of identified Prakriti type, the proposed system came up with personal recommendations on lifestyle, such as dietary, physical activity, yoga, and daily routine practices in accordance with the Ayurvedic principles. These recommendations were dynamically adjusted to the specific dosha type and real-time physiological state of the user by the recommendation engine.

Fig. 5. Dosha Prediction and Recommendation.

Besides on-screen suggestions, the system automatically created a detailed individual health report in PDF format. All the relevant products of the system, such as the classified Prakriti (dosha type), registered physiological parameters,

ECG wave- form characteristics, heart rate, oxygen saturation (SpO_2), and pulse rate are condensed in this report. The PDF report offers a concise and organized report on the health condition of the user by offering both diagnostic information and preventive recommendations in a single piece of paper.

Evaluation in terms of the user level showed that the recommendations that were generated were context-aware and easy to interpret. The fact that a downloadable PDF report was available further improved the usability because it allowed keeping the records long-term and sharing it easily with healthcare experts.

D. Comparative Analysis

The proposed system, in comparison to traditional IoT-centered health mon- itoring systems which limit their functionality to tracking parameters, provides a higher-functioning proposal, including constitution-based analysis and preventive wellness advice. The proposed method is less subjective as it uses real- time physiological data unlike the questionnaire-based Prakriti identification methods. The system is more comprehensive in healthcare as it shows increased customization, comprehensive evaluation, and prevention concerns.

E. Result and Discussion

The experimental results demonstrate that combining IoT technology and machine learning with Ayurvedic principles is a viable path for personalized healthcare. While the system performs exceptionally well with dominant Prakriti types, identifying mixed Doshas remains a complex task with room for further optimization. Our pilot validation confirms that the hardware-software pipeline can accurately distinguish physiological signatures, such as the rapid heart rates (exceeding 90 bpm) seen in Vata types compared to the slower rhythms (approximately 60 bpm) of Kapha types.

TABLE III
SUBSET OF REAL-TIME PHYSIOLOGICAL VALIDATION DATA

Subj. ID	Pulse Rate (bpm)[a]	Pulse Press. (mmHg)[b]	Temp. (°C)	Actual Dosha	Pred. Dosha
S01	71.20	63.83	30.99	Pitta	Pitta
S02	75.48	64.94	30.95	Pitta	Pitta
S03	60.63	72.44	28.74	Kapha	Kapha
S04	92.90	37.06	28.92	Vata	Vata
S05	93.63	31.99	28.86	Vata	Vata
S06	62.15	70.80	28.55	Kapha	Kapha

[a] Represents the Frequency of the heart beat (Speed).
[b] Represents the Amplitude or strength of the pulse wave (Volume).
Pulse Pressure: Low values indicate a "Thready/Weak" pulse (Vata), while High values indicate a "Strong/Full" pulse (Kapha).

However, factors like sensor precision and dataset scale continue to influence overall performance, highlighting the need for larger and more diverse training sets. A key strength of the current framework is its ability to move beyond simple classification by generating detailed, Nadi-Pariksha-style wellness reports. By bridging the gap between raw biological signals and actionable lifestyle advice, the system offers a more explainable and user-centric approach to preventive medicine.

F. Automated Wellness Reporting

Unlike traditional classification models that output a single label, the proposed system generates explainable wellness reports similar to industry-standard Nadi Pariksha devices. By correlating predicted Dosha types with the user's specific sensor readings, the platform produces a 'Prakriti Card' detailing personalized diet, lifestyle, and yoga recommendations. This feature bridges the gap between automated digital diagnosis and actionable expert advice.

Fig. 6. Sample Automated Prakriti Card and Wellness Report

VI. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The paper has introduced Prakriti, a combined IoT and Machine Learning -based healthcare platform that was meant to identify human body constitution (Prakriti) and suggest personalized lifestyle changes based on the Ayurvedic principles.

The system operates based on IoT sensors to track the vi- tal physiological parameters of such parameters as heart rate, ECG, body temperature, and oxygen saturation (SpO_2) continuously. These real-time cues are learned with supervised machine learning methods, that is, Ensemble Learning models on Dosha classification and Support Vector Machines (SVM) on the early detection of health risks. The proposed approach was feasible and reliable as it was demonstrated by the experimental assessment with high levels of classification accuracy and low latency and the ability to maintain stable performance.

In contrast to traditional healthcare monitoring systems that are based more on analyzing diseases and symptoms, Prakriti puts more emphasis on preventative and holistic health through analyzing the biomedical data using the Ayurvedic Tridosha system. Robotic determination of body composition does not require any manual diagnosed expert diagnosis, and individual diet, exercise and yoga recommendations enable consumers to make wise lifestyle decisions in advance.

B. Future Scope

Although the existing system depicts good performance there are various avenues that can be enhanced in the future:

- **Mobile Application Integration:** The creation of native An- droid and iOS applications, as an alternative to the web-based dashboard, with offline data caching and push notifications.
- **Deep Learning Implementation:** Adding to it Deep Learning models at a more granular level, including Long Short- Term Memory (LSTM) networks to dissect temporal patterns of ECG and pulse data to predict the heart in more detail.
- **Enhanced Data Security:** Incorporating Blockchain technology to provide hashing and data safety in storing sensitive health records of patients, and this overcomes privacy limitation in the cloud based IoT services.
- **Clinical Validation:** Enlarging the sample by partnering with Ayurvedic hospitals to test the model recommendation on a broader group of patients.

REFERENCES

[1] S. Kande and R. Ingle, "A Review on IoT Based Smart Healthcare Monitoring Systems," *Int. J. Eng. Res. Technol.*, vol. 9, no. 6, 2020.

[2] P. Sindhuri, M. Yuvaraj, C. Pradeep, M. S. Vishnushree, and V. Thamaraiselvan, "Personalized Health Recommendation System with Machine Learning," *Int. J. Res. Publ. Rev.*, 2025.

[3] M. Raja, M. Lakshman, M. A. Teja, M. Vinay, and M. V. Vardhan, "Personalised Healthcare Web-Application," *Int. J. Innov. Sci. Res. Technol.*, 2025.

[4] K. Parvateesam et al., "Design and Implementation of Advanced Real-Time Patient Health Monitoring System Using IoT," *Int. J. Eng. Res. Technol.*, 2024.

[5] M. Alarfaj et al., "Wearable Sensors Based on Artificial Intelligence Models for Human Activity Recognition," *Front. Artif. Intell.*, 2024.

[6] B. Yadav et al., "IoT-Enabled Health Monitoring Systems: Transforming Healthcare Delivery," *Int. J. Res. Appl. Sci. Eng. Technol.*, 2024.

[7] B. G. Mohammed and D. S. Hasan, "Smart Healthcare Monitoring System Using IoT," *Int. J. Interact. Mob. Technol.*, 2023.

[8] P. Bidve, S. Mishra, and A. J., "Enhancing Ayurvedic Diagnosis Using Multinomial Naïve Bayes and K-Modes Clustering," 2023.

[9] S. K. Jagatheesaperumal et al., "An IoT-Based Framework for Personalized Health Assessment and Recommendations Using Machine Learning," 2023.

[10] S. M. Sharrath et al., "Patient Health Monitoring System Using IoT," *Int. J. Creat. Res. Thoughts*, 2022.

[11] F. Sabry et al., "Machine Learning for Healthcare Wearable Devices: The Big Picture," *J. Healthc. Eng.*, 2022.

[12] H. A. I. Madusankha et al., "Exploring the Ayurvedic Concept of Body Types and Their Relevance to Heart Rate Variability," 2021.

[13] P. Tiwari et al., "Recapitulation of Ayurveda Constitution Types by Machine Learning of Phenotypic Traits," *PLOS ONE*, 2017.

[14] A. J. Joshi, "Acquisition and Quantitative Analysis of the Arterial Pulse," Ph.D. thesis, Indian Inst. Technol. Bombay, Mumbai, India, 2011.

[15] A. Joshi et al., "Nadi Tarangini: A Pulse Based Diagnostic System," in *Proc. IEEE Eng. Med. Biol. Soc. Conf.*, 2007.

Acknowledgements

We make a move to offer our genuine thanks to **Dr. Mahesh Pawaskar** for his important direction in this task. We are especially appreciative for his untiring help with this project as well as for empowering us in taking out the mistake and kept our advancement on plan. We are likewise thankful for the co-activity and significant help delivered by **Dr. Jaya Gupta** as Head of the Department. We might want to offer our unique gratitude to the next instructing and non-showing staff of the AIML Department for their valuable help in the finish of this task.

At long last, we are enormously obliged to the administration of our school and express our appreciation to the Principal **Dr. Uttam Kolekar** for organizing and giving essential office in finishing this venture work. We wish to recognize the assistance gave by our kindred schoolmates to aiding the assets while finishing this undertaking. We truly thank all other people who have assisted us in this venture with working.

Suraj Vishwakarma 22106019

Shravan Thakur 22106086

Sameer Singh 22106116

Ritesh Singh 22106037

Date: